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# Technical Report No. 88

## Microbial Data Deviation Investigations in the Pharmaceutical Industry

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## Technical Report No. 88

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# 1.0 Introduction

Microbial data deviations are microbial test results that fall outside the product specification or acceptance criterion established in a drug application, drug master file, official compendia, good manufacturing practice regulations, or internally by the manufacturer. This includes, but is not limited to, results generated during environmental and water monitoring as well as pharmaceutical ingredient, in-process, and finished product testing. The validity of the test results is supported by a documented laboratory and manufacturing investigation, where applicable.

The information presented in this technical report was compiled by a team of experts representing both industry microbiologists and regulators and reflects current industry best practices. The membership of personnel from the U.S. Food and Drug Administration (FDA) on the task force does not imply Agency endorsement. Feedback from industry and regulatory professionals was evaluated and incorporated into the technical report to ensure that the suggested methods, terminology, and practices for conducting microbial data deviation investigations reflect sound science and current technological approaches. This technical report is not intended to establish any mandatory or implied standards. Published references supporting the recommendations presented are cited in the text and listed in **Section 7.0 References**.

Definitions for the terms used throughout this report are provided in the Glossary. When available, the same or similar technical definitions have been used as those included in the FDA or International Conference on Harmonization (ICH) guidelines. To avoid ambiguity, the task force members believe that use and definitions of these terms should also be consistent and standardized when incorporated into company-specific standard operating procedures, validation protocols, and regulatory submissions.

## 1.1 Purpose

The FDA issued a *Guidance for Industry: Investigating Out of Specification (OOS) Test Results for Pharmaceutical Production* in 2006 in response to the U.S. versus Barr Laboratories decision, which is applicable to laboratory testing results for products regulated by the Center for Drug Evaluation and Research (CDER) ***(1, 2)***. The FDA guidance does not extend to microbiological failure investigations, however. Consequently, PDA developed this technical report to provide approaches, specific to the pharmaceutical industry, for investigating various types of microbial data deviations based on current scientific knowledge and modern testing technologies. The term "microbial data deviation" is preferred to "out-of-specification" or "out-of-limit result" in this technical report, as most microbiological tests, especially in a sterile product manufacturing facility, are in-process tests, not finished- product tests that must meet a regulatory-approved specification. For simplicity and consistency, the term "deviation" will be used throughout this document.

Technical Report No. 88 presents a holistic approach for performing a microbiological investigation. It provides a framework to assist with focusing on the investigational areas that may contain or contribute to the root cause of data deviations, such as sampling, test methodology, and suitability of the test, as opposed to providing methods for analysis of batch acceptability. (Batch acceptability determinations and release decisions are made by the quality control unit as required by FDA good manufacturing practice (GMP) regulations, not by deviation investigation teams.)

## 1.2 Scope

Each of the three sections of this document focuses on distinct areas where microbial data is collected and inspected for deviation investigations and with the following designated responsibilities:

**Roles and Responsibilities**—Focuses on personnel, especially microbiologists, in the laboratory investigations and cross-functional teams responsible for manufacturing investigations.

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**Conducting Laboratory Investigations (Phase I)**—Provides additional details on how an investigation may be conducted to determine that the microbial test was valid and what is the root cause of a failure when the result is deemed invalid. Subsections focus on different microbiological tests conducted, with points for consideration in the final determination of the investigation.

**Conducting Manufacturing Investigations (Phase II)**—Illustrates where a microbiologist, acting as a subject matter expert (SME), can contribute to the evaluation of personnel activities, environmental changes, utilities, materials, and process-related events and can assist in determining the probable cause of the deviation and the most appropriate corrective and preventive actions (CAPA).

If needed and when appropriate, Phase I and Phase II investigation activities can be conducted in parallel.

The roles of other contributors to deviation investigations required for effective root cause analysis, such as representatives from manufacturing, process engineering, and quality assurance, will also be briefly described. The composition of the investigation team will depend on the nature of the investigation.

This technical report promotes a lifecycle approach, but greater emphasis is placed on laboratory and manufacturing investigations of marketed drug products. The approaches described may be suitable during research and product development and the manufacture of clinical supplies, as well as the manufacture and testing of regulatory-approved products. The microbial tests discussed in this technical report include microbial attribute testing of nonsterile products, components, sterility testing, bacterial endotoxin assay, mycoplasma testing, environmental and personnel monitoring, and microbial monitoring of utilities including pharmaceutical-grade water and compressed gases.

Microbial assays of antibiotics and vitamins and the detection and enumeration of protozoans, viruses, and prions are not included in the scope of this report. Additionally, batch disposition is separate from the outcome of a microbial investigation report. This document is directed specifically to pharmaceutical manufacturing; however, the industry practices may be suitable for deviation investigations in the biological, consumer health, medical devices, and cosmetic industries. In addition, these can be extended to contract manufacturing and testing organizations supporting pharmaceutical companies.

While this document provides a framework illustrating how deviations and the resulting investigations might be conducted, each company is responsible for developing their internal policies and procedures suitable for its specific needs that also meets current GMPs.

# 2.0 Glossary of Terms

### Action Limit

A microbial or total airborne particulate limit that, when exceeded, indicates a process is outside of its normal operating range. A response to such an excursion should involve a documented investigation and corrective actions based on the results of that investigation *(3-6)*.

### Alert Level

An established value that when exceeded, provides an early warning of potential drift from normal operating conditions and validated state, which does not necessarily give grounds for corrective action but triggers appropriate scrutiny and follow-up to address the potential problem *(3,4,7)*.

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### Adverse Trend
An increase in the frequency of alert level or action limit excursions or repeated recovery of low levels of microorganisms below the alert level during microbial monitoring or of pharmaceutical ingredient or finished product failure that is indicative of a loss of process control.

### Batch Record Review
Review by the quality control unit of the batch manufacturing record for accuracy and completeness and for absence of derivation from the approved manufacturing and testing processes. The batch record is inclusive of all in-process testing and release records for a batch.

### Continuous Improvement
Ongoing activities to evaluate and positively change products, processes, and the quality system to increase effectiveness.

### Correction
Repair, rework, or adjustment relating to the disposition of an existing deviation.

### Corrective Action
Action taken to eliminate the causes of an existing deviation or other undesirable situation to prevent recurrence.

### Corrective and Preventive Actions (CAPA)
A subsystem used to collect and analyze information, identify and investigate product and quality problems, and take appropriate and effective measures to prevent recurrence of the identified problem ***(8)***.

### Cross-Functional Team
An investigatory team, comprising representatives including QCU microbiologists, manufacturing, engineering, and maintenance personnel, and other appropriate SMEs, with the purpose of conducting a manufacturing investigation.

### Deviation
Data or a result outside of the expected range or an unfulfilled requirement; it may be called nonconformity, defect, discrepancy, out-of-specification, out-of-limit, or adverse trend.

### Drug Substance
Active pharmaceutical ingredient in a drug product responsible for its therapeutic activity.

### Drug Product
A finished dosage form, for example, tablet, capsule, or solution, that contains a drug substance, generally, but not necessarily, in association with one or more other ingredients ***(9)***.

### Environmental Monitoring Excursion
Monitoring for nonviable particulates and/or microorganisms where the result meets or exceeds the alert level or action limit.

### Excipients
Inactive pharmaceutical ingredients in a drug product responsible for its manufacturability and physicochemical attributes.

### Impact Analysis
Determination of how a product failure may extend an investigation "to other batches of the same drug product and other drug products that may have been associated with the specific failure or discrepancy" ***(10)***.

### Invalid Result
Laboratory test result confirmed to be invalid as determined by a laboratory investigation.

### Invalid Test
Laboratory test that, as a result of the laboratory (Phase I) investigation, did not meet the test method requirements and whose results would not be deemed valid. This may also apply to a test which was aborted (e.g. breakdown of isolator during sterility testing).

### Error, Laboratory
A self-evident documented mistake that will bring the validity of a laboratory test into question.

### Error, Manufacturing
A self-evident documented mistake that will bring the manufacturing process into question.

### Microbial Data Deviation
A microbial test result that deviates from approved specifications or in-process limits.

### Out-of-Level Result (Out-of-Limit Result)
Any unexplained deviation or result that failed to meet set levels for environmental monitoring and other utilities.


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### Out-of-Specification Result
Any unexplained deviation or the failure of a pharmaceutical ingredient, drug substance, or drug product batch to meet any of its regulatory-release and shelf-life specifications.

### Quality Control Unit
An organizational element within a pharmaceutical company with the authority and responsibility for quality as defined by 21 CFR 211.22 ***(11)***. The quality control unit typically comprises the quality control and quality assurance functions.

### Quality Risk Management
A systematic process for the assessment, control, communication, and review of risks to the quality of the drug (medicinal) product across the product lifecycle.

### Recall
An action taken by a firm to remove a product from the market. Recalls may be conducted through a firm's own initiative, by health authority request, or by health authority order under their statutory authority.

### Right First Time
The percentage of the time that a manufacturing process step, batch record, or in-process or laboratory test is successfully completed on the first attempt.

### Risk
The combination of the probability of the occurrence of harm and the severity of that harm to the recipient of a drug product.

### Risk Assessment
A systematic process of organizing information to support decisions to be made within a risk management process consisting of the identification of hazards and the analysis and evaluation of risks associated with exposure to those hazards.

### Specifications
A list of tests, references to analytical procedures, and appropriate acceptance criteria that are numerical limits, ranges, or other criteria for the test described. It establishes the set of criteria to which a material should conform to be considered acceptable for its intended use.

## 2.1 Abbreviations/Acronyms

**BET** Bacterial Endotoxin Test

**CAPA** Corrective and Preventive Actions

**DMF** Drug Master File

**EM** Environmental Monitoring

**OOL** Out-of-Level/Out-of-Limit

**OOS** Out-of-Specification

**PPE** Personnel Protection Equipment

**QA** Quality Assurance

**QC** Quality Control

**QCU** Quality Control Unit

**SME** Subject Matter Expert

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# 3.0 Microbiologist Roles and Responsibilities

## 3.1 Laboratory Microbiologist Responsibilities

The principal responsibility for achieving accurate laboratory testing results resides with the analyst performing the microbiological test. Recognition of potential practices that could create invalid results should be an integral part of the laboratory analyst's training. The analyst is responsible for performing all testing according to standard operating procedures (SOP) and in-house documentation practices. The analyst should only use qualified, calibrated, and well-maintained laboratory instruments and equipment and approved growth media, diluents, and reagents.

The analyst should check the product test results for compliance with their specifications before discarding test preparations, whenever practical. If any potential or suspected laboratory errors occurred during testing, the analyst must discontinue testing immediately and notify their laboratory management. The analyst, with guidance and instruction from the laboratory management, must collect and retain all test samples, reagents, consumables, paperwork, and any other relevant materials related to the affected assay, even if the test is deemed invalid, to aid the laboratory investigation. If errors are clear and indisputable, such as atypical appearance of a test sample (e.g., unexpected particulate matter present), damaged test container, test sample spillage, use of wrong pipetting volumes, missing dilution steps, aseptic handling missteps, or use of the wrong testing medium, the analyst should immediately document the event. When in doubt, the analyst may consult with the laboratory management to confirm that the test should be discontinued.

The analyst should not knowingly continue a test that they expect to invalidate at a later time for an assignable cause (i.e., analyses should not be completed for the sole purpose of seeing that a passing result can be obtained when obvious errors are known). In addition, the analyst is responsible for assisting management with the laboratory investigation.

## 3.2 Microbiologist Role in a Manufacturing Investigation

The key to successful manufacturing investigations of microbial testing failures with pharmaceutical ingredients and finished drug products is the inclusion of an experienced microbiologist as an SME on the manufacturing investigation team. This microbiologist may work in the quality control (QC) microbiology testing laboratory at the manufacturing site, the quality control unit (QCU), product development, or a corporate microbiology support group. Ideally, the microbiologist should have a comprehensive knowledge of product formulation and specific manufacturing operations, obtained from frequent shop-floor visits or surveillance of the overall process. For smaller organizations, without internal microbiological expertise, the microbiologist may be an outside consultant. To be truly effective, the SME must be a trained microbiologist with broad experience in microbiological investigations. Furthermore, it would be advantageous if the SME is familiar with the actual product implicated in the failure. Without this expertise on the investigation team, it is less likely that the root cause of the failure can be determined, and the investigation may be hampered or severely limited.

The microbiologist should be engaged in the investigation as soon as is practical, as microbiological investigations are always time-sensitive.

The microbiologist should be part of the first step in an investigation, such as performing a walk-through of the manufacturing area, to become oriented and identify situations and events that are nonroutine and may be directly linked to the deviation. To be successful, the microbiologist must have access to all related testing results, have the cooperation of manufacturing managers, supervisors, operators, and support personnel, and have the authority to interview staff, inspect processing equipment, take investigatory microbial samples, and review change control and batch records. In some instances, access to customer complaint records may be necessary. For failures related to purchased pharmaceutical ingredients, it is highly desirable that a microbiologist join any audit team visiting the ingredient supplier to conduct an investigational review.

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Systematic, proactive risk assessments for assessing the overall aseptic manufacturing process and controls have been proven helpful in identifying potential weaknesses in advance, and may mitigate the occurrence of a deviation, for example, sterility failures and endotoxin or environmental monitoring (EM) excursions ***(12)***.

## 3.3 Microbiology Laboratory Management Responsibilities

Upon notification by a laboratory analyst of a suspected laboratory error, the laboratory management must conduct an objective assessment to determine if an error has occurred. Data should be reviewed promptly to ensure the contemporaneous memory of events and to determine if the results might be attributed to an isolated or systematic laboratory error. In the absence of laboratory error, a confirmed valid test could indicate problems in the manufacturing operations; however, all facts need to be evaluated before predetermining a cause. The correct compliance assumption is that the results are valid until proven otherwise.

Laboratory management should ensure that the probable cause is supported by evidence, using root cause analysis tools when appropriate (e.g., 5 Whys, Ishikawa/fishbone diagram). Laboratory management should promptly notify the QCU and any impacted departments of the suspected deviation and inform them that a laboratory investigation has been initiated and when it might be completed.

If testing is conducted by a contract testing laboratory, the quality agreement between the contract testing laboratory and the manufacturer should clearly assign responsibilities, include a timeline, and specify the requirements of the laboratory investigation.

## 3.4 Subject Matter Expert Responsibilities

SMEs can support the investigation by providing technical assistance related to the assay methods and technologies utilized, designing any investigational testing, and providing general scientific guidance, as needed. This expert could have greater experience and scientific background related to the assay or manufacturing process from within the company. Hiring a qualified outside consultant may also be an option.

## 3.5 Quality Control Unit Responsibilities

The QCU oversees all aspects of the microbial deviation investigation. A QCU representative or other approved personnel would be responsible for coordinating with the cross-functional team involved in supporting the investigation, including personnel from manufacturing, facilities, calibration, validation; approving any laboratory retesting plans; and approving the laboratory and manufacturing investigations. The QCU also would be responsible for overseeing quality agreements made with contract testing laboratories and for ensuring that microbial deviation investigations are adequately addressed.

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# 4.0 Phase I: Conducting the Microbiology Laboratory Investigation

The first phase of a microbial data deviation (deviation) investigation is an initial assessment of the results reported by the microbiology laboratory. The purpose of the laboratory investigation is to establish the validity of test results, thereby confirming that a deviation has occurred. If a laboratory error has occurred, which would invalidate the test, the QCU would approve retesting and/or resampling of the pharmaceutical ingredient or drug product/drug substance. Some compendial microbiology testing chapters have limited information to address how to handle sample deviations and investigations ***(13-14)***. This technical report will discuss additional factors for microbiological evaluation of sterility, bacterial endotoxin, and other microbiological testing conditions for invalidating test results or allowing an initial test to be repeated.

Pharmaceutical QC microbiology laboratories vary in size and complexity. These factors may impact the level of experience with dealing with deviations in terms of the training, experience, and academic background of the analysts and laboratory management.

Contrary to the belief that microbial failure investigations are usually inconclusive, experience has demonstrated that the causes of the microbial contamination are determinable through robust investigations conducted by teams that include an experienced microbiologist. In the absence of a definitive cause, the most probable cause(s) can be determined based on the available information and follow-up observations, additional testing, and batch-record review. To avoid a confirmation bias, the team must be willing to reject their original hypothesis about the probable cause in light of the evidence subsequently collected. If possible, as many questions should be asked as to why the probable cause is not correct as are asked about why it is correct. Hindsight, where the successes of past microbial failure investigations often are exaggerated and may incorrectly influence the direction of the current investigation, is another bias that must be recognized and avoided. However, a history of conducting investigations will enhance the success with current investigations.

When no obvious laboratory errors are noted, the laboratory management must initiate a laboratory investigation promptly and must complete it as soon as is practically possible in accordance with regulatory expectations and company policies and procedures. Accurately recording the sequence of activities may prove helpful in determining root cause during an investigation. Some tests, such as microbial identification, may require more time and may delay completion of the laboratory investigation. **Figure 4.0-1** presents a generalized fishbone diagram to be used as guidance for how to evaluate the inputs of microbial testing during root cause analysis.

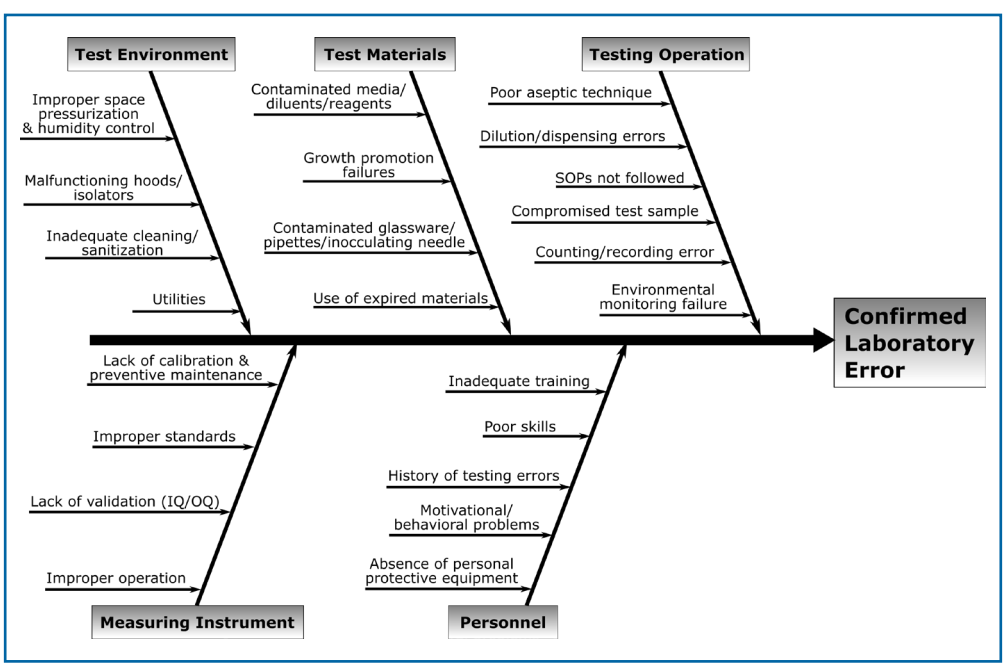


**Figure 4.0-1** Fishbone Diagram Showing Inputs of Microbial Testing during Root Cause Analysis


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**Table 4.0-1** lists sample questions that may be asked during the laboratory investigation. This suggested list of questions and conditions for consideration should not be considered all-inclusive.

**Table 4.0-1** Sample Questions for Laboratory Investigations

| Laboratory Process or Component | Questions for Consideration |
|---|---|
| Sampling and/or Shipment to the Microbiology Laboratory | Were the samples collected using proper aseptic technique? |
| | Were the samples taken in a sampling cabinet (where applicable) so not to contaminate the test material? |
| | Were sterile sampling devices used (e.g., sterile scoops, sampling thieves)? |
| | Was the sampler gowned appropriately to protect the sample from contamination? |
| | Were personnel properly trained to take samples? |
| | Were personnel wearing appropriate personal protective equipment (PPE)? |
| | Were sample containers intact (not damaged or opened) and sterile prior to testing? |
| | Was the sampling area properly sanitized, using qualified sanitizing agents? |
| | Did errors occur during routing, delivery, and storage of the test samples to the laboratory (e.g., dropped, opened, spilled, and/or improperly stored)? |
| | Were all procedures followed and confirmed (e.g., container-closure integrity testing, label review and verification, attribute inspection) during sample collection? |
| | Did the chain of custody and the time lapse between sampling and testing adhere to procedural requirements? |
| | Were the samples transported in a secure manner that did not contribute to microbial contamination (e.g., single-use protective barriers, properly sanitized reusable tote)? |
| | Were the associated documentation and sampling forms properly completed with all relevant information necessary for testing? |
| | Were the samples transferred into the laboratory promptly and stored under proper conditions prior to use in testing? |
| Testing Procedures | Was the test method performed in accordance with the current standard operating procedure? |
| | Is the test method correct and appropriate for the sample being tested? |
| | Was method suitability testing performed on the applicable sample type? |
| | Were the samples tested in a controlled environment using proper aseptic technique? |
| | Is the analyst properly qualified, experienced in running the test, and current with their training? |
| | Is the procedure and/or test method clear, or could it be subject to interpretation? |
| | Was the analyst properly gowned, wearing appropriate PPE for the test (e.g., face mask, sterile sleeves, sterile gloves, etc.)? |
| | Were the appropriate materials, media, diluents, and components used for the test? |
| | Were the testing consumables accepted into inventory via a defined incoming-materials process and within the appropriate expiration date? |
| | Did the materials, media, and diluents pass quality control testing, where required? |
| | Was the lot information for the materials, media, and diluents properly documented? |
| | Were any changes made to how reusable glassware were cleaned and sterilized (e.g., new employee, new detergent, new washer, new autoclave)? |
| | Was there a trainee or new analyst working with the lead analyst? |
| | Were the required incubation time and temperature used, if applicable? |
| | Was water of the appropriate quality and volume used for media and solution preparation? |
| | Was the testing area properly sanitized, using qualified sanitizing agents? Were all agents within their expiration date? |

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(Table 4.0-1 Continued)

| Laboratory Process or Component | Questions for Consideration |
|---|---|
| Testing Procedures | Was the sanitization of the testing and support areas performed properly and as frequently as required by the current laboratory procedures? |
| | Were the environmental and/or personnel monitoring samples collected as part of the testing session appropriate and acceptable? |
| | Were all system controls (manipulative) properly executed and acceptable? |
| | Were all negative controls for media and diluents properly executed and acceptable? |
| Equipment/Supplies | Has the microbiology laboratory experienced any unusual events that could compromise the laboratory testing areas (e.g., construction, open doors, standing water, temperature excursions, spills, HVAC failure)? |
| | Were the equipment, instruments, and pipette aids used in the test properly maintained and calibrated? |
| | Were related, reusable, testing equipment (e.g., micropipettes, pipet-aids), glassware, and materials properly cleaned and/or sterilized prior to use, according to procedure? |
| | Were all the pieces of equipment and supplies brought into the testing area properly handled? |
| | Were the gowning components properly cleaned and/or sterilized, and was the packaging fully integral? |
| | Did any maintenance of equipment, planned or unplanned, occur that could have had a potential impact on the test or result? |
| | Was a water bath or sonicator required for testing and, if so, was it properly cleaned, sanitized, and filled with the appropriate quality of water? |
| | Was the container or packaging used in holding microbiological culture media integral (i.e., devoid of cracks or missing components)? |
| | Were all the analysts trained on the proper use the equipment? |
| Data Collection, Recording, and Calculation | Was any anomalous or suspect information, aside from that being investigated, recorded in the raw test data? |
| | Did any microbiological media, aside from that being investigated, exhibit microbial counts and/or turbidity? Were all containers and plates from the deviation test session reexamined? |
| | Were all calculations used to convert raw data into final test results scientifically sound, appropriate, and correct? |
| | Was a review of historical data conducted for any reagents and media involved in a past deviation? |
| | Did all media and equipment used for the test meet quality control specifications (for growth promotion, sterility, pH, appearance, incubation, sterilization, etc.)? |
| | Was a review of all EM data conducted for the area where the sample was tested, as well as for the adjacent areas (e.g., transfer rooms, incubators, etc.)? |
| | Was the test method performed as prescribed by the qualified method? |
| | Has laboratory management reviewed all the raw testing data to determine if any correctable laboratory errors have occurred (e.g., clerical, mathematical, transcriptional errors)? |
| Microbial Identification in Support of the Investigation | Was a Gram stain performed (i.e., review slides under a microscope to confirm the Gram reaction and cellular morphology and identify all of the microorganisms recovered during the analysis)? |
| | Is the microbial identification appropriate for the isolate (e.g., genotypic, phenotypic, proteotypic)? |
| | Were the microbial identification platform(s) appropriately qualified according to either the manufacturer's instructions or a recognized procedure, such as that described in *USP <1113> Microbial Characterization, Identification, and Strain Typing?* ***(16)*** |
| | Were any commercially available cultures and/or facility isolates used as controls? Were they appropriately maintained, according to procedure? |
| | Based on the criticality of the test (e.g., sterility test), was the identification of the microorganism(s) implicated in the investigation confirmed via strain-typing or molecular fingerprinting? |


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If a laboratory error (Phase I investigation) cannot be found that invalidates the results, the original test results are presumed to be correct. The next step, then, is to conduct a Phase II investigation, including review of manufacturing batch records and other facility documents available to find the possible cause of the deviation or trend that may have occurred during manufacturing. Specific points to consider might include:

- If an analyst thinks a mistake has been made in any part of the testing, including sample preparation, the assay should be terminated, and the justification documented.

- Analysts should be instructed to neither destroy nor discard the sample, sample-preparation dilution tubes, or reagents, when possible, until the final results are known, confirmed, and released. Comparing them with the product specification may be an important part of the laboratory investigation.

- Caution should be used in conducting investigational assays. They are not considered retests, but a response to the deviation and should not be used to release material. A laboratory SOP should precisely describe the procedure(s) that must be followed if a deviation is observed. Such a document eliminates unplanned decisions regarding the appropriateness of investigational testing, retesting, or resampling. When appropriate, retesting can be conducted according to established compendial directives.

- Investigational testing provides the laboratory an opportunity to assess the hypotheses regarding possible reagent or analytical failures in testing; it frequently aims to identify and change a variable (e.g., analyst) from the first failed test. Investigational testing should be well-defined and justified, preapproved, and well-documented. Values generated from the investigational test may provide a clue as to the actual or potential product contamination or a mishandling of a positive control or a tube mix-up. The purpose of performing investigational tests is not to reverse the original deviation, but to gain information regarding a potential failure, so a more effective effort can be made to pinpoint the root cause and implement any necessary CAPAs.

Fully documenting each step in the lab investigation is critical. Laboratory management should determine not only the reliability of the individual test result obtained, but also the significance of the deviation. If the deviation is verified to be correct, they must determine what that represents about the laboratory quality assurance program and what impact the final report may have on other product lots. Laboratory management should continue to monitor for developing trends.

General points to consider during the investigation phase include:

- Having all QC challenge microorganisms genotypically and phenotypically profiled may be advantageous so a library is available to compare product isolates in the event of cross-contamination during testing.
- If growth is detected on the negative control media, these isolates should be identified for investigative purposes.
- The two-fold variance described in USP <61> and <1111> as the maximum acceptable count applies only to approved specifications in cfu/g or mL, expressed as a power of 10 ***(15, 17)***.
- Nonsterile products tested at one or two product dilutions for bioburden (e.g., 1:10, 1:100) may need to be retested at higher product dilutions to obtain a countable number of colonies, if the highest product dilution tested yields a result too numerous to count.
- Due to the variability of microbiological results, the investigation should not be limited to the specific batch but extended to multiple batches, excipients, or APIs over a selected period of time.
- Include historical trend data for the assay, controls, and any additional factors that may support the investigation hypothesis and/or investigational results.

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## 4.1 Microbial Identification

Identifying the microorganism(s) implicated in product failure is a critical step in a laboratory investigation and, subsequently, in the manufacturing investigation. Knowledge of the source of the microorganism may lead to a possible route to determine which microorganism contaminated the product.

Microbiological identification systems are based on different analytical methodologies, with their inherent limitations, and the databases related to the system. Identification is accomplished by matching characteristics (genotypic, phenotypic, or proteotypic) to an established standard (reference) microorganism, such as a strain-type in the database. If a microorganism is not included in the library or database of the system, it will not be identified. Therefore, pharmaceutical manufacturers should review the breadth of the database of the identification system they plan to use and its applicability to their use requirements.

As stated in *USP <1113> Microbial Characterization, Identification and Strain Typing*, the hierarchy of microbial identification errors of impact, in descending order, is (1) misidentification to genera, (2) misidentification to species, and (3) no identification ***(16)***. Incorrect identifications could lead to a faulty investigation, inappropriate CAPAs, and ill-advised product disposition. For example, a misidentification may result in a nonsterile drug product being released when it contains an objectionable microorganism. If the in-house identification system cannot identify the isolate or if there is doubt regarding the identification, it should be sent to a contract testing laboratory for confirmation; otherwise, a misidentified microorganism will always be carried in the records.

The need for microbial identification is specifically cited in *USP <62> Microbiological Examination of Nonsterile Products: Tests for Specified Microorganisms*. This chapter presents a requirement for confirmatory identification tests for organisms that grow on selective or diagnostic media and demonstrate defined morphological characteristics ***(18)***. As a related aspect, *USP <71> Sterility Tests* allows for invalidation of the test if, after identification of the microorganism(s) isolated from the test, the contamination may be unequivocally ascribed to faults with respect to the material and/or the technique used in conducting the sterility test procedure (discussed further in **Section 4.2.1**) ***(13)***.

Taking a polyphasic identification approach is recommended, that is, one that assembles and assimilates many levels of information from molecular, physiological, morphological, serological, or ecological sources to classify a microorganism. Some vendors of identification instruments encourage microbiologists to forgo the preliminary characterization steps, such as colony description, Gram reaction, cellular morphology description, and/or diagnostic screening. Forgoing these steps is not recommended as it often necessary to assess the microorganism identification against all available information to determine if the identification makes sense. For example, *Salmonella typhi* is unlikely to be isolated from a cleanroom, but this pathogen could be present in an animal-derived ingredient.

Another consideration is that many phenotypic identification systems require the preliminary characterization of the isolate for the selection of the appropriate identification kit. An incorrect Gram stain may result in a misidentification and flawed investigation. Criticality of this routine test should not be overlooked.

**Table 4.1-1** outlines general points to consider when encountering a microorganism associated with a product failure.


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**Table 4.1-1** General Points to Consider in Microbial Identification

| Identification Aspect | Considerations |
|---|---|
| Instrumentation and Data | Is the instrumentation used for the identification adequately maintained and within calibration? Did the reagents pass their quality control checks? |
| | Was the probability of the match to the database sufficiently high or are supplemental tests required? |
| | Would a small change in the results lead to a completely different identification? Does the isolate need further subculture for full phenotypic expression? |
| | Is the isolate difficult to differentiate from other related species with the identification technology employed, e.g., members of the genus *Bacillus*? |
| Microorganism Characteristics | Was there a past history of isolation of this species in your manufacturing facility, utilities, pharmaceutical ingredients, or drug products? |
| | Do the microbial characteristics and physiological requirements make it likely that the microorganism could persist in the manufacturing facility, utilities, pharmaceutical ingredients, and/or drug products? |
| | Is the microorganism likely to be isolated from people or from a facility in the geographic or climatic zone the facility is located? |
| | Is the microorganism associated with a rare disease? Have fastidious growth requirements made it an unlikely identification? |

## 4.2 Phase I: Specific Points to Consider

### 4.2.1 Sterility Testing Investigations

If a laboratory sterility test result used for release testing of a pharmaceutical or biological drug product or a medical device indicates microbial growth in the microbiological growth medium (i.e., turbidity, pellicle formation, precipitation, or flocculent growth) during the incubation period and is subsequently confirmed via subculture and/or Gram reaction, it raises the question: Was the observed growth a result of a sampling error, laboratory error, or a true product contamination? This scenario with a finished product is commonly referred to as an 'out of specification' (OOS) test result.' Due to the unique nature of viable microorganisms, unlike chemical analytes, microbiologists also describe these aberrant results as a 'microbiological data deviation' (MDD) The response to this presumptive failure to meet USP <71> requirements is to investigate the source of the microbial contamination in order to confirm the validity of the test results. The purpose of the investigation is to confirm that the test results were obtained while utilizing good laboratory practices, including proper aseptic technique, appropriate equipment and reagents, and sterile assay media/components within an appropriate Grade A/ISO 5 environment.

Deviations in equipment, procedures, or personnel performance that require corrective action should be discovered during the investigation. If the positive sterility test is determined to be valid, an investigation will be conducted to determine the root cause of the manufacturing practice that caused the tested product lot to become contaminated. If no definitive root cause is determined, then the manufacturing corrective actions might be based on the probable causes identified and the measures defined to prevent recurrence of such an event. The decision on the disposition of the batch needs to consider that, as no analytical cause was identified, the batch might be nonsterile. If a laboratory investigation concludes that errors or events caused the test contamination to come from the laboratory environment or met other compendial invalidation criteria, the sterility test result would be invalidated, and the sterility test would be repeated with the same number of units and quantity of product.

USP <71> requires that the sterility test media be visually examined for evidence of microbial growth at intervals during the incubation and at its conclusion. This evidence may present itself as turbidity throughout the media, a pellicle growing on the surface of the soybean-casein digest medium (SCDM), growth at the interface between the aerobic and anaerobic portions of the fluid thioglycollate medium (FTM), as floccular growth within the media, or as precipitation at the bottom of the media. The mode of growth should be documented within the sterility test report.

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Although not specifically recommended in USP <71>, microscopic examination of suspected microbial growth and subculturing the contents of test media container/enclosed membrane canisters are common industry practice. The sampling of the media must be done aseptically to avoid additional microbial contamination. Incubation of the original container must complete the full 14-day period described in USP <71> in order to ensure discovery of slower growing microorganisms. Aliquot removal from the canisters (or sterility test vessel used for direct inoculation) can be challenging when using a sterile Pasteur pipette (or sterile syringe) as it is difficult to assure the same subsequent culture growth conditions that may be needed in the event of an investigation. For example, the proximity of the growth within the media may be diagnostic as to the type of microorganism(s) that had been recovered. Secondly, it is not advisable to disperse the macroscopic growth within the media, especially for FTM because the aeration of the media may be fatal for the presence of strict anaerobes.

Subculturing to isolate the suspected microbial contaminant may include:

- Transferring up to 10 mL of the test media into fresh SCDM and FTM and incubating the media for at least 14 days at 20-25 °C and 30-35 °C, respectively. The subculture time of incubation should mimic the time period that was necessary for the original time of incubation in order to optimize microbial recovery required by the targeted species (for example, some environmental fungi require the lower temperature range and are inhibited by the higher temperature range).
- Streaking out from the test media onto soybean-casein digest agar (SCDA) and Sabouraud dextrose agar (SDA) and incubating aerobically at 30-35 °C for SCDA and 20-25 °C for SDA, until microbial growth is obtained and/or,
- Alternative enrichment medium (i.e., Columbia blood agar) may also be used for subculturing from the primary SCDM and FTM to support slow growing fastidious microorganism.

The medium and mode of growth within the medium will provide insight as to whether the microorganism is a strict anaerobe, facultative anaerobe, or strict aerobe and what subculturing method would be most appropriate. Any isolates should be identified at least to species and, if available, strain-typed to aid the follow-up investigations with isolates recovered from the manufacturing or material bioburden. If subculturing fails to recover the microorganism, then a direct Gram stain on an aliquot of the suspect microbial growth should be performed. If no microbial presence is detected with the direct Gram stain, the turbidity may be a result of product-media precipitation or an artifact of the media after prolonged incubation (for example, the low percent of agar in the FTM can precipitate out of solution and causes medium turbidity).

Careful review and documentation of the microbial appearance within the SCDM and the FTM test tubes or canisters before any inadvertent jostling or shaking This is especially true when the product may potentially make the test media turbid or cause a precipitation (e.g., suspensions or liposome-based drug products). These observations should be recorded in the sterility test report or, if available, photomicrographs can be taken with a camera mounted on the microscope.

In the absence of the isolation of a microorganism on subculture of the test media and/or no microscopic evidence of the presence of microorganisms, the deviation sterility test failure is not confirmed and, therefore, the product passes the test.

USP <71>, which has been harmonized with the *European Pharmacopoeia* and *Japanese Pharmacopoeia,* is frequently used as the method of choice for finished product release and is cited in many regulatory applications ***(19, 20)***. The section entitled "Observation and Interpretation of Results" describes the four acceptable conditions that may be used to invalidate the initial test result and allow for retesting:

1. The data of the microbiological monitoring of the sterility testing facility show a fault (**Table 4.2.1-1**)
2. A review of the testing procedure used during the test in question reveals a fault (**Table 4.2.1-2**)
3. Microbial growth is found in the negative controls (**Section 4.2.1.1**)

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4. After determination of the identity of the microorganisms isolated from the test, the growth of this species (or these species) may be ascribed unequivocally to faults with respect to the material and/or the technique used in conducting the sterility test procedure (**Section 4.2.1.2**)

If evidence of microbial growth is found, the product to be examined does not comply with the test for sterility unless it can be clearly demonstrated that the test was invalid for causes unrelated to the product to be examined. If the test is declared to be invalid, it is repeated with the same number of units as in the original test. If no evidence of microbial growth is found in the repeat test, the product examined complies with the test for sterility. If microbial growth is found in the repeat test, the product examined does not comply with the test for sterility ***(14)***.

During the laboratory phase of the investigation, the review process should be formalized in a standard operating procedure or other type of Work instruction document. To be consistent with the USP criteria, the investigative review questions in this section have been grouped in the same order.

**Table 4.2.1-1** Data of the Sterility Testing Facility Show a Fault

| Identification Aspect | Considerations |
|---|---|
| Testing Facilities | Has the sterility testing isolator system, glovebox, or laminar flow hood been certified to meet the required certification standards? |
| | Has any of the test equipment (e.g., filtration apparatus, filter) been associated with other recent sterility issues? |
| | Has the isolator chamber, glovebox, or laminar flow hood been properly decontaminated by chemical disinfection, e.g., vapor phase hydrogen peroxide? Has the decontamination process been successfully validated? |
| | Are there any leaks in the gloves? (Review the glove integrity test and EM data including fingertip glove samples) |
| | Has the sterility testing suite been disinfected according to laboratory SOPs? Were there any recent changes to the disinfection solutions used before the appearance of the suspect test results? |
| | Has the cleanroom HEPA filtration certification been performed as scheduled? Does it meet requirements for air cleanliness, air velocity, space pressurization, temperature, and humidity? |
| | Were the room conditions acceptable (e.g., temperature, differential pressure, humidity)? |
| Testing Method | How vulnerable is the sterility test method (e.g., direct inoculation, enclosed membrane filtration methods) to contamination? Are there risk mitigation practices in place? |
| | Is the sterility test method conducted in an isolator, glovebox, or laminar flow hood? |
| | Has the disinfecting solution been prepared according to labeling requirements? |
| | Do the post-sanitation microbiological monitoring results meet acceptable criteria? |
| | Has the microorganism isolated from the product test results appeared in routine EM (air, surface, and personnel monitoring) in the sterility test facility? |
| | Were the external surfaces of the test sample disinfected with a sporicidal agent prior to bringing the sample into the aseptic testing environment? |

### 4.2.1.1 Microbial Growth Found in Negative Controls

If the appearance of microbial growth is found in the negative or manipulation control (i.e., system control), they should be carefully reviewed for the presence of contamination. Not all microbial contamination causes uniform media turbidity (e.g., surface film layer on top of the media for strict aerobes or small budding fungal growth attached to membrane filter or walls of glass containers). Any purchased media, diluent, and test equipment should be checked for manufacturing recalls. Alternatively, in-house-prepared media records should be reviewed for contamination trends.

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**Table 4.2.1-2** Testing Procedure Used Reveals a Fault

| Fault Aspect | Considerations |
|---|---|
| Testing Materials | Did all sterility test media used pass QC checks (pH, appearance, integrity of container) and relevant growth promotion and sterility checks? |
| | Were the expiry and storage conditions of the sterility test media confirmed? |
| | Does the product have a history of sterility failure issues? |
| | Were the Certificates of Analysis for the commercially available media and equipment used for the analyses (canisters, preformulated media, diluents) accurate and acceptable? |
| | Were the product and media containers and any other supplies properly disinfected before being placed into the laminar flow hood or isolator? |
| | Does any of the suspected glassware contain product contamination (e.g., appearance of residual material within the glassware from a prior sample or other contamination source)? |
| | Was the container-closure integrity of the product confirmed? |
| | Do the original test plates used for isolation show possible preexisting contamination (i.e., growth in non-streaked locations on the agar surface or subsurface growth)? |
| | Were cleaning and sterilization requirements for reusable glassware and equipment properly implemented? |
| | Do the QC records show proper sterilization of all equipment and media used during the sterility test method (e.g., manifold, rinse fluid, culture media, forceps, scissors)? |
| | Were the samples used for any other type of nondestructive QC testing prior to their arrival at the sterility test area? |
| Testing Method | Were any unusual occurrences or accidents during the sterility test reported by the analyst(s)? |
| | Are the training records and qualifications of the analysts performing the test up to date? |
| | Was an appropriate method suitability test performed for the product in question? |
| | Was the same microbial identification method for environmental isolates from sterility testing used as for the isolate(s) in the sterility-positive unit(s)?<br>**Note:** Different microbial identification technologies and/or databases may generate different results. |
| Testing Environment | Was the area used for subculturing onto isolation media after the growth observation in the sterility-test medium adequate to prevent adventitious contamination that would compromise the integrity of the identity of the sterility test isolate? |
| | Was the laboratory area workspace cluttered? Were there unsanitized surfaces that may have caused plate contamination? |
| | Was there any growth on the exposed plates used within the working hood (or isolator) during the time of sample analysis or any growth from personnel samples? |
| | Was sufficient EM performed to allow detection of low levels of microbial contamination that may be present? |
| | Are the media and incubation conditions for the samples adequate to detect contamination?<br>**Note:** The EM plate results only have significance if microbes are recovered; no recovery does not assure asepsis of the sterility-test work environment. |

### 4.2.1.2 Decision Based on Microorganism Identification

EM data recorded during the sterility test should be reviewed to determine if the test isolate matches a microorganism recovered from any environmental or personnel monitoring plates. In the event of a species match, it may be necessary to perform strain-typing, for example, partial ribosomal RNA sequencing or riboprinting, in order to attribute the microorganism associated with the failure to the EM isolate from the sterility-testing suite.

If the test failure microorganism is the same genus or species as an organism used by the laboratory for growth promotion or suitability testing, (i.e., QC challenge organisms), conducting a genotypic profile on both isolates would be advisable. When making this comparison, USP <1113> may be consulted.


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#### 4.2.1.3 CAPAs to Address Invalidation of a Sterility Test

The firm is responsible for developing suitable, case-specific CAPAs if a legitimate cause to invalidate the sterility test has been documented (as introduced in **Section 4.2.1)**. Potential corrective action steps that could be implemented or strengthened for remediation of a problem associated within each of the four conditions outlined in USP <71> include:

1. **Data of the microbiological monitoring of the sterility testing facility show a fault**
   - Establish a documented QA/QC program to ensure the sterility and growth promotion capabilities of media, diluents, materials, and equipment used
   - Employ a sporicidal agent for the sanitization/disinfection of the area concerned, and continue to use the sporicidal agent periodically
   - Retrain personnel in the GMP of aseptic technique and the performance of the operations
   - Establish or strengthen a scheduled preventive maintenance and certification program for cleanroom equipment, and increase the microbial monitoring of the personnel and critical areas to assure the environmental controls
   - Validate and monitor (using data from each sterilization cycle) the autoclave processes used to sterilize cleanroom equipment

2. **A review of the testing procedure used during the test in question reveals a fault**
   - Establish appropriate SOPs for media preparation, storage, inventory control, and destruction of expired media
   - Employ a requirement for two operators to conduct sterility testing to minimize fatigue while optimizing aseptic handling and for action verification
   - Establish a separate designated area for all subculturing activity, and prepare an SOP for cleaning and monitoring this laboratory area

3. **Microbial growth is found in the negative controls**
   - When a negative control is found positive, require a second person, e.g., peer or laboratory manager, to observe the contamination
   - Check the container-closure integrity of the test container as it may have been damaged during manufacturing, storage, or transportation to the microbiology laboratory
   - Review component sterilization, material transfer, environmental controls, and aseptic technique used during testing.

4. **After determination of the identity of the microorganisms isolated from the test, the growth of this species (or these species) may be ascribed unequivocally to faults with respect to the material and or the technique used in conducting the sterility test procedure.**
   - When comparing the isolates from the test failure to personnel and EM, past sterility, and media-fill failures, use the same or equivalent identification technologies to confirm the identification of the isolates from the failed test.

### 4.2.2 Bacterial Endotoxins Testing Investigations

Compendial bacterial endotoxins testing (BET) detect and/or quantify lipopolysaccharide cell membrane components derived from the outer-cell membrane of Gram-negative bacteria that are most commonly responsible for human pyrogenic responses. This method is more comparable to chemical assay than to the traditional methods used for the detection and enumeration of microorganisms in pharmaceuticals, biopharmaceuticals, and medical devices. For this reason, the enzyme cascade-based *Limulus* amoebocyte lysate (LAL) BET has unique analytical requirements that may affect the reliability of the test results. USP <85> *Bacterial Endotoxin Tests* describes the procedures for performing both the gel clot and spectrophotometric methods for detecting and quantifying bacterial endotoxins. Because the methods are based on enzymatic components and principles, incubation conditions (e.g., temperature and time) are important, and labora-

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tory analysts should have a technical understanding of the potential inhibition and enhancement of the product matrix on the product/lysate mixture. Another critical factor is the measurement and, if necessary, the adjustment of the product–lysate mixture to fall within the required pH range as stated in the lysate instructions for use. If the pH is not within the specified range, the endotoxin test results can be significantly altered from the true results (usually lower). Therefore, it is important that analysts understand the details of the compendial test and are trained to recognize how changes in the procedures may cause erroneous results is imperative ***(14)***. **Section 4.2.2** describes general approaches for investigating BET deviations and provides guidance and references that contain detailed protocols on how to conduct investigations.

A deviation may be due to one of the following from a partial list of reasons:

- Error during product dilution (excessive volume transfer during serial dilution preparation), incorrect volume (too little or too much) during reagent rehydration, misplacement of handled test tubes causing a control standard endotoxin (CSE) serial dilution for positive control preparation and inhibition and enhancement results to be incorrect, mislabeling of test tubes, or a test tube mix-up between product and CSE serial dilutions
- CSE cross-contamination caused by pipette tips (or pipets) used repeatedly
- Automated pipettors or bulbs contaminated with aspirated endotoxin solution because of improper use or not being cleaned routinely
- Working on a lab bench with a nearby splash sink that creates water aerosols (tap water commonly contaminated with Gram-negative bacteria)
- Analyst not performing the test according to the prescribed method
- Reagent(s) or equipment not adequately depyrogenated or not handled aseptically with pyrogen-free water
- Unexpected occurrence during the analysis that interrupted the completion of the gel clot (e.g., table vibration from draw closure or fan vibration in air-circulating stand-up incubators)
- Glucans, at high enough concentrations, present in the test material may cause false positives
- Contaminated water baths used for incubation of gel clot test tubes

**Section 4.2.2.1** compiles a number of factors important to BET that, if not performed correctly, may culminate in a faulty analysis.

#### 4.2.2.1 Compendial Bacterial Endotoxins Test

When faced with an endotoxin LAL OOS result it would be best to start with a review of the mandatory requirements and controls in the harmonized compendial chapter, USP <85>. As with any testing used to release products, performing these key analytical steps and documenting them accurately and contemporaneously as required by company SOPs and the lysate manufacturer's instructions for use is essential. These records will prove critical in the event the laboratory management or auditors review or question the validity of the results. The harmonized compendial chapters include specific considerations for addressing BET analytical anomalies:

- In the section for INTERPRETATION of the harmonized gel-clot method, when a positive result is found for only one duplicate of Solution A (product only), the test should be repeated. In the repeat test, the preparation under test complies with the test if a negative result is found for both replicates of Solution A. The preparation does not comply with the test if a positive result is found for one or both replicates of Solution A. However, if the preparation does not comply with the test at a dilution less than the maximum valid dilution (MVD), the test may be repeated using a greater product dilution, not exceeding the MVD.
- Along with each gel-clot Limit and Quantitative Test, a 2 λ positive product control, a 2 λ positive control in BET water, and a negative control with just BET water should be included.
- For the Photometric Quantitative Techniques, the absolute value of the correlation coefficient, r, must be 0.980 for the range of endotoxin concentrations used.
- For the Photometric Quantitative Techniques, to be considered free of factors that interfere with the assay under the conditions of the test, the measured concentration of the endotoxin added

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to the sample solution must be within 50%–200% of the known added endotoxin concentration after subtraction of any endotoxin detected in the solution without added endotoxin. **Note:** Knowing how to interpret the value of the positive product control (PPC) is very important when reading either under or over the allowed range. Even an assay within the allowable PPC range may still give false test results, which may be caused by the natural endotoxin in the test sample and the spiked CSE combining to give an acceptable value for this control.

- Additional suggestions on how to investigate bacterial endotoxin deviation can be found in *USP <1085> Guidelines on the Endotoxins Test*, in the section entitled "Out-of-Specification Results and Retesting Considerations" ***(21)***.
- For those laboratories responsible for performing BET on sterile implantable devices or combination products (one that includes a medical device), specific guidance is provided in *USP <161> Medical Devices–Bacterial Endotoxin and Pyrogen Tests* ***(22)***. For medical device testing, careful attention to the compendial requirements should be given to performing the bacterial endotoxin extraction step from the device products.

#### 4.2.2.2 FDA Guidance

When performing a deviation investigation for results associated with BET, the 2012 *FDA Guidance for Industry: Pyrogen and Endotoxin Testing: Questions and Answers* may offer valuable insights as to some regulatory expectations ***(23)***. In general, the guidance suggests that the sampling plan used by firms should consider the potential presence of bacterial endotoxin contamination in raw materials, in-process materials, and the finished product. Specifically, they should consider aspects of the manufacturing design, including consistency of a manufacturing process, impact of in-process hold-times, endotoxin-removal steps, and finished-product endotoxin specifications. Sections of this FDA guidance document may be particularly helpful during deviation investigations, for example:

> **When is retesting appropriate?** When conflicting results occur within a test run, firms should consult USP Chapter <85>, Gel Clot Limits Test, Interpretation, for guidance on repeat testing. As specified in Chapter <85>, if the test failure occurred at less than the maximum valid dilution (MVD), the test should be repeated using a greater (product) dilution not exceeding the MVD… All testing procedures, including those for retesting within the above limits, should be specified in advance in the standard operating procedures approved by the firm's quality control unit.

Because of the unique nature of endotoxins to clump or bind to container surfaces, the issue of sample storage and handling becomes important. The FDA guidance recommends firms "establish procedures for storing and handling (which includes product-mixing) samples for bacterial endotoxin analysis…."

> FDA recommends that pooled samples be a composite of aseptically removed aliquots (after at least 30 seconds of vigorous mixing) from each of the product containers ***(24)***. In this way, the original, individual containers will be available for possible retesting in the event the pooled sample displays an OOS result.
>
> …For purposes of evaluating the relative risk of product contamination, quantitative testing may be preferable to limit testing to facilitate product quality trending and to identify and correct excursions before they exceed the specification and cause product failure.

Therefore, when an endotoxin is detected, quantitating the actual amount present is preferred to depending on the guesswork of using a limit test approach.

Two other documents that may be helpful during the investigation of either a laboratory deviation or the manufacturing areas are: FDA's *Inspection Technical Guides: No. 40 Bacterial Endotoxins/Pyrogens* and standards document *ANSI/AAMI ST72:2019 Bacterial Endotoxins—Test Methodologies, Routine Monitoring, and Alternatives to Batch Testing* ***(25, 26)***. The AAMI standard is acknowledged as an acceptable

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standard for performing BET and has been referenced in the 2012 FDA Guidance document excerpted above. Both of these documents may be used by FDA and other international government investigators during their inspectional preparation of both parenteral and medical device manufacturers.

### 4.2.3 Antimicrobial Effectiveness Testing Investigations

The test methods and acceptance criteria for different dosage forms are defined in *USP <51> Antimicrobial Effectiveness Testing*. In general, the test is only applicable to aqueous, multiple-use sterile and nonsterile drug products that may be subject to microbial contamination during use. USP <51> defines aqueous as a water activity ($A_w$) of more than 0.6. Products with $A_w < 0.6$ will not support the growth of microorganisms and may be demonstrably self-preserving. The test confirms the effectiveness of the antimicrobial preservative system and/or other supporting physicochemical attributes to meet product category criteria. Although a drug product would be required to meet the test criteria during development and to be demonstrated as effective in a long-term stability study, the test is a shelf-life specification, but not a release requirement. In contrast, the preservative concentration required within the formulation to meet the product category criteria at release and at expiry is a product specification included in the regulatory submission. Typically, USP <51> is looked upon as a stability-indicating pass/fail test, where the results in terms of log reduction of the challenge organisms at specified time intervals are evaluated against the log reduction requirement for the product category. Given the inherent variability of the test method, marginally preserved products have a potential to fail at one or more time intervals in long-term stability studies supporting marketed products ***(27)***.

#### 4.2.3.1 Investigating Antimicrobial Effectiveness Test Deviation

**Table 4.2.3.1-1** outlines points to consider in a laboratory investigation of an antimicrobial effectiveness test deviation**.**

**Table 4.2.3.1-1** Questions Concerning Antimicrobial Effectiveness Testing

| AET Component | Considerations |
|---|---|
| Sample Preparation | Was the challenge organism inoculum of the proper quality and purity? |
| | Was the appropriate inoculum concentration used? |
| | Was the proper qualified diluent used in the inoculum preparation? |
| | Was the inoculated product well mixed prior to storage and at the time of testing? |
| | Was the pH of the product within the specified range? |
| | Was the preservative concentration as measured by chemical assay within specification, e.g., 80-120% of label claim? |
| Incubation | Were the test materials held at 20-25 °C during the 28-day test period? |
| | Were the stability samples incubated upright or inverted? |
| Results Evaluation | For evaluation of the statistical robustness of the number of colonies per pour plate over a 10-fold dilution range: Did the replicate plates fall into the range of 25 to 250 colonies per plate for bacteria and 8 to 80 for mold ?<br>Are the counts on the plates comparable? |
| | Was the growth-promotion testing conducted with the appropriate microbiology growth medium? |
| | Was the challenge organism recovery method validated? |
| | Is the recovered growth morphologically consistent with the inoculum organism? |
| | Was the laboratory worksheet complete and the log reduction calculations accurate (i.e., were appropriate rounding rules followed?)? |

### 4.2.4 Mycoplasma Testing Investigations

Mycoplasma contamination is a serious concern when using mammalian, plant, and insect cell cultures in the development and production of cell-derived biopharmaceutical products for human use. These products may include viral vaccines, monoclonal antibodies, growth factors, replacement thera-

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pies, and gene and cell therapies. Contamination of cell culture ingredients and cell lines represent a potential patient safety risk as well as an economical loss in terms of reduced product yields, rejections, and recalls. To minimize these risks, master and working cell lines or banks, cell culture ingredients, inoculum expansions, process materials, and finished products are routinely screened for mycoplasma. The most common contaminating mycoplasma species (95%), in descending order, are *Mycoplasma orale* (human cell lines), *M. hyorhinus* (porcine cell lines), *M. arginini* (bovine cell lines), *M. fermentans* (human cell lines), *M. hominis* (human cell lines), and *Acholeplasma laidlawii* (bovine cell lines).

#### 4.2.4.1 Laboratory Investigation for Out-of-Specification Result

The mycoplasma contamination rates within cell cultures, especially cell banks, historically have been high ***(28,29)***, though screening and removal strategies, such as irradiation and heat treatment, have progressively reduced the rates. When the mycoplasma testing laboratory has a failing result, how does the laboratory management and QCU know that it is a valid result?

Points to consider when conducting the laboratory investigation of culture- or PCR-based detection methods are presented in **Tables 4.3.4.1-1** and **4.3.4.1-2**, respectively.

**Table 4.3.4.1-1** Culture-Based Mycoplasma Testing Considerations

| Culture Testing Component | Considerations |
|---|---|
| Sample/Organism | What mycoplasma species was detected in the test specimen? |
| | Could that species be found on laboratory personnel or in the cell line? |
| | Is there a history of presence of mycoplasma in the specific cell culture, cell culture media, processing ingredients, upstream processing, or finished product? |
| | What was the test sample (e.g., fresh, frozen, or freeze-dried cell pellets; genomic extracts from the cell line; cell culture media ingredients; mammalian cell harvest for viral vaccine product)? |
| | Did the test sample contain low or high levels of mycoplasma contamination as determined on the duration of the enrichment steps to detect mycoplasma?<br>**Note:** Contaminated cell lines will contain higher mycoplasma counts, whereas media ingredients that have mycoplasma inactivation steps and no cells to support mycoplasma growth may have low-level mycoplasma contamination. |
| | Although cell culture media will not show turbidity or a pH change, do cell cultures of the contaminated cell line show morphological signs of mycoplasma contamination (i.e., cytopathogenic effects or presence of fluorochrome-stainable mycoplasma)? |
| Method Suitability | Did the test meet the USP <63> method suitability requirements?***(29)*** |
| | Did the media pass the growth promotion tests? Did the positive and negative controls give the required results? |
| | Was *Mycoplasma orale* recovered from a non-human mammalian, microbial, insect, or plant cell line? If yes, it may have been introduced during upstream processing or testing. As it is the most common mycoplasma species in the oral cavity of healthy people, it has the greatest potential as a testing contaminant. |
| | Has the analyst conducting the test been adequately trained and is experienced with the test? |
| | Was the test conducted with adequate personnel and environmental controls? |

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**Table 4.3.4.1-2** Mycoplasma PCR-Based Detection Testing Considerations

| PCR Testing Component | Considerations |
|---|---|
| Reagents/Sample Preparation | Was there evidence that the reagents, i.e., master and assay mixes, were satisfactory? Did the inhibition, negative, and positive controls meet the specified requirements? Were the amplification plot, melting temperature, and derivative value interpreted correctly? |
| | Was the test environment adequately isolated? Is the test method an open or closed system? |
| Method | Was the method validation and suitability testing for the test sample adequate? |
| | Was the initial test repeated in duplicate with two of three replicates being clearly positive? |
| Confirmation | Does the follow-up culture method (required with PCR failures) support or dispute the initial PCR-based assay reported failure? |

### 4.2.5 Biological Indicators Investigations

Biological indictors (BI) are used during the validation of moist heat, dry heat, and gaseous sterilization processes and during vapor-phase hydrogen peroxide decontamination cycles. BIs may also be used for monitoring routine cycles to support the release of products that are used by the pharmaceutical industry. Additionally, a BI can be used as a load monitor to support parametric release of moist-heat sterilized products in the U.S. Typically, they consist of high numbers of resistant bacterial spores with characterized calibrated D-values that are included in sterilization loads, retrieved, and cultured to demonstrate the lethality of their destruction during the sterilization cycle.

It is an expectation that a company will conduct confirmatory testing of the population and purity stated in the certificate of analysis of the incoming BI lot and growth of the cultured spore strip. Low counts may reflect the incomplete recovery from the matrix supporting the bacterial spores, whereas the growth of a heat-resistant spore-forming bacteria like *Geobacillus stearothermophilus* from a BI is most likely caused by contamination of the laminar flow hood and/or test supplies from positive controls in the testing environment ***(30)***.

Probable causes of failure would include gross mishandling, insufficient exposure to the sterilization parameters, and higher BI D-values that result in a lower probability of destruction in the sterilization cycle. Other points to consider include:

- Were the BIs stored and handled prior to use, as recommended by the manufacturer?
- Were the BIs incubated in accordance with regulatory expectations and manufacturer instructions?
- Were the BIs used in the manner prescribed by the manufacturer?
- Was the appropriate BI used for the product being tested (e.g., ampoule BI used for liquid products)?
- In cases where the BI is directly inoculated into a solution or onto a substrate, what is the D-value as used in that form?

PDA Technical Report 51 Biological Indicators for Gas and Vapor-Phase Decontamination Processes: Specification, Manufacture, Control and Use provides further guidance on this topic ***(31)***.

### 4.2.6 Aseptic Process Simulation Failures (Media Fills)

Proper design and execution of aseptic process simulations can be found in PDA Technical Report 22: *Process Simulation for Aseptically Filled Products*, FDA *Guidance for Industry: Sterile Drug Products Produced by Aseptic Processing—Current Good Manufacturing Practice*, PIC/S *Annex 1: Manufacture of Sterile Medicinal Products*, *Guide to GMP for Medicinal Products*, and ISO 13408-5 ***(3, 5, 32, 33)***. Any media-filled containers that, upon inspection, score as "turbid" during the incubation period should be examined by a microbiologist and transferred to the microbiology laboratory for investigation. The growth of microorganisms in culture medium may include any slight haze, pellicle formation, floccular amassment, or precipitation. The first task, prior to opening the container to subculture the media, is to establish if the container is flawed in any manner that would compromise the sterility of the medium. Damage may take the form of inadequate crimps to a stoppered vial, leakage around the container closure, or chips or cracks in the container that

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could threaten integrity. Any damage should be adequately recorded and, if possible, supplemented with photographs. Involving packaging experts in the laboratory investigation who can evaluate the extent of damage may be helpful, for instance, in performing physicochemical integrity testing to confirm that the damage would result in microbial ingress. The media fill may be inspected for damaged/nonintegral units, and those units should be discarded. The rationale for excluding these units from the final run tally should be fully documented and the deviation should be explained in the media fill report. ***(3, 34)***.

In the absence of obvious damage, the contents of a turbid container would be aseptically sampled and subcultured with appropriate solid media and incubated at 20–25 °C and 30–35 °C for three to seven days. The contents of the media fill containers may be examined microscopically for the presence of microorganisms. Any isolates should be identified at least to species and, preferably, strain-typed to aid the follow-up investigations. Guidance may be found in USP <1113> ***(16)***.

### 4.2.7 Environmental Monitoring

Laboratory investigations are an opportunity to take stock of the sampling program and operation of the microbiology laboratory. Common questions that may be asked during a laboratory investigation include those in **Table 4.3.7-1.**

**Table 4.3.7-1** Environmental Monitoring: Questions Concerning Laboratory and Sampling

| Laboratory Component | Considerations |
|---|---|
| Sample Collection | Was the correct sampling method/technique performed? |
| | Was the sampling technician properly trained and qualified? |
| | Was the sampling technician properly gowned for the area requirements? |
| | Were the negative controls handled properly? Were results negative? |
| | Was the correct swab or contact plate used? |
| Media Handling | Was the media appropriate for intended monitoring (i.e., tryptic soy agar, Gamma-irradiated media, double- or triple-wrapped media)? |
| | Was the media properly received and stored in good condition? |
| | Was the media growth promoted? Were the results passed, approved? |
| | Was the lot used within expiry as supported by growth-promotion studies? |
| | Were the swabs, sterile diluent, and liquid media within expiry? |
| Equipment | Was the active air sampling equipment used within the current calibration period? |
| | Was the equipment functioning properly during sampling? |
| | Were the equipment batteries charged? |
| | Were the sampling heads appropriately sterilized or sanitized prior to use? |
| | Was the equipment handled properly and not dropped during use? |
| | Was the appropriate flow rate delivered to the sample? |
| Media Incubation | Did incubation take place at the correct time and temperature? |
| | Was the incubator temperature within the acceptable range throughout the incubation? |
| | Did any equipment alarms or power outages occur during incubation? |
| | Were the samples handled with care and not dropped? |
| Microbial Identification | Was the colony color and morphology documented? |
| | Was a Gram stain performed and cellular morphology determined? |
| | Were any preliminary biochemical reactions performed? |
| | Were microbial identifications conducted correctly and documented? |
| | Was the organism reviewed against routine EM results to determine if it had been seen at lower levels of contamination prior to the event? |
| | Was the source of the organism (e.g., water, personnel) determined to evaluate where the contamination came from? |

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### 4.2.8 Utilities Monitoring

#### 4.2.8.1 Pharmaceutical-Grade Water Systems

The purpose of the laboratory investigation is to determine whether a definitive source of sampling and/or laboratory error contributed to the root cause of the deviation. This section addresses the initial and extended laboratory investigations associated with samples collected from pharmaceutical-grade water systems, such as purified water and water for injection (WFI).

Microbiological testing of pharmaceutical water systems, including the source water (i.e., potable water) and purification steps, typically consists of total viable microbial count, BET, and/or a test for the absence of coliforms, depending on the grade of water and any local regulatory requirements. After a pharmaceutical water system is successfully validated, both the system and the source are subject to periodic microbial monitoring using standard methods. Action limits and alert levels should be established to recognize adverse trends in the water system. These alert levels or action limits should be based on historical data, the intended purpose of the sampling plan, as well as the recommended compendial or regulatory limits. In general, specifications are not set for water systems.

In a well-designed, well-operated pharmaceutical water system, sanitized by maintaining a minimum turbulence level and elevated temperature (> 65 °C) or periodic heat or chemical sanitization, test results above the action limit should be very rare.

Several events can signal that a laboratory investigation should be conducted:

- A single test result that exceeds an action limit
- Multiple test results that exceed an alert level, thus indicating an adverse trend
- Isolation of microorganisms of concern based on the dosage forms manufactured in the facility

The cross-functional team conducting the laboratory investigation should review the collection and microbiological testing of the subject water sample to determine any potential sampling and laboratory errors. The data collected should include the sampling location, sample containers, sampling technique, and any alert level or action limit excursions associated with the sampling location, loop, or water system, and the operators collecting the water samples.

If this investigation does not identify a definitive source of laboratory error to explain the deviation result, the laboratory should immediately notify the owner of the utility system, operational users, and the QCU.

##### 4.2.8.1.1 Evaluation of Sample Handling

The methods used for preparing, storing, and attaching hoses and sampling apparatus to the sample ports are critical. Regulatory guidance recommends that each point of use should be flushed using the same flushing procedure used in the manufacturing process ***(35)***.

The potential for changes in the microbial population in samples should be limited by restricting the holding time before testing and controlling the storage temperature. If the samples cannot be tested within two hours, they should be transported and stored at refrigeration temperature for a maximum of 24 hours. In situations where even 24 hours is not possible (such as when using off-site contract laboratories), it is particularly important to qualify the microbiological sample hold times and storage conditions. (See APHA/AWWA/WEF *Standard Methods*, USP <1231>, EMA 496873, and Japan: G8 for guidance ***(28, 36-38)***. Prior to removing the test aliquot, all samples should be adequately mixed to ensure a homogeneous mixture is achieved. Any parameters for storage and transportation of water samples that are different from standard methods should be qualified to ensure the samples are not negatively affected, resulting in potentially invalid results.


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#### 4.2.8.1.2 Evaluation of Test Methods

Review of all materials associated with the testing and their appropriateness for the grade of water tested is required in a laboratory investigation. Sterility of the membrane filters, sample containers, pipettes, and utensils used to perform testing must be suitable for their intended use and must conform within sterilization dating. Also, all diluents and reagents used must be suitable (such as sterile and/or endotoxin-free) and within their expiry dating.

Any deviation from these requirements must be evaluated and investigated. Confluent microbial growth, in contrast to discrete colonies on the membranes or pour plates, may be indicative of equipment contamination. The position of the analyst's head, hands, and arms in the path of the airflow (preferably in a laminar flow hood with horizontal air flow) over the test materials may also be a source of microbial contamination (personnel-related considerations are discussed further in **Section 5.2.1**).

The identity of the microorganism(s) associated with the investigation may also reveal information about the source of contamination. The most common water isolates are Gram-negative, nonfermenting, rod-shaped bacteria historically described as pseudomonads, which may be indicators for a biofilm formation within the water system. The recovery of organisms that do not thrive or survive under the low ionic strength/low carbon conditions found in a pharmaceutical water system may suggest contamination of the sample or insufficient control or flushing of the sampling apparatus and/or hoses attached at the port. Gram-positive cocci commonly found on the human skin, such as *Staphylococcus epidermidis*, can be indicative of potential sampling contamination, while Gram-positive sporeformers, such as *Bacillus* species or the fungus *Cladosporium* species, can be indicative of potential environmental contamination.

#### 4.2.8.1.3 Evaluation of Trending Data

Reviewing the historical test data will help determine if an adverse trend is associated with individual points-of-use, loops, and water systems. This information will be also useful for the manufacturing investigation.

## 4.3 Phase I: Conclusion and Next Steps

Evaluation and questions related to the Phase I deviation investigation must use strong, evidence-based data. The investigation may not be definitive but may suggest probable cause(s). The Phase I investigation commonly concludes with one of three possible outcomes:

1. Events uncovered during the investigation may be responsible for the deviation and present a clear root cause that is directly responsible for the deviation. Examples could show contamination due to failure to follow SOPs, use of poor aseptic techniques during either sampling or testing, a recognized laboratory error, or the use of nonsterile testing materials. There may be a direct link between isolate identification and the same contamination from the testing environment. In these cases, the test may be considered invalid.

   In the case of invalid tests, the Phase I investigation would conclude without advancing to the Phase II manufacturing investigation, or where Phase I and Phase II investigations were undertaken concurrently, the cancellation of the Phase II investigation would be warranted. Depending on the sample type and source of contamination, resampling or retesting may also be considered as long as the QCU agrees. CAPAs must be implemented (preferably prior to any retesting or resampling), however, in order to remediate the original cause of the invalidation.

2. While the investigation might have revealed issues, there may be insufficient evidence to directly correlate the investigative findings to the deviation. For example, if the EM conducted during the testing showed microorganisms present but the organisms were a different genus and species than the contamination within the sample, the environmental results may cast doubt on the sample, but they

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would not invalidate the original result. Thus, the Phase I laboratory investigation would be found inconclusive, not providing sufficient evidence to invalidate the result, and the result is deemed to be valid. The investigation would then proceed to a Phase II manufacturing investigation.

3. The results of the investigation did not show any evidence of testing failure. In other words, it can be clearly demonstrated that all procedures were followed, analysts were properly qualified to perform the testing, and all materials, equipment, and testing environment were considered appropriate for the test. The calculations for reporting data and all final results were correct and verified. Therefore, the deviation is confirmed and deemed valid, so the investigation must proceed to Phase II in order to determine the source of contamination from the processes used during manufacturing.

# 5.0 Phase II: Manufacturing Investigation

Once laboratory error has been eliminated or found inconclusive through the Phase I investigative process, the investigation focuses on the manufacturing facility, personnel, processes, and materials. A manufacturing investigation includes, but is not limited to, several principal elements:

- The cross-functional investigation team must represent all departments responsible for handling the deviated sample whether it is a pharmaceutical excipient, in-process intermediate, drug substance, or drug product. This component is essential to conducting an effective investigation. The QCU or another suitable individual may lead the team, which should include representatives from manufacturing, plant engineering, process validation, maintenance, and quality control. In a microbiological-related investigation, the contribution of a microbiologist on the team will be critical.

- The QCU should have overall responsibility for managing or approving the outcome of the manufacturing investigation and for notifying upper management of the investigation findings and batch disposition.

- Each investigation should be thoroughly documented and include a time for completion, as specified by company policies and procedures. Microbiological-related investigations are complex and may not be able to be completed within the specified time period; in these cases, an SOP-governed mechanism should be in place for extending the completion date. This extension must be justified and supported by the QCU.

While additional assessments can be performed if required by company policy or procedure, every manufacturing investigation should include the following actions:

- **Determine all batches in scope.** Does the initial scope of the investigation need to be expanded based on laboratory findings? Batches under investigation should not be released.
- **Determine the definitive or probable cause (root cause analysis).** The investigation must be comprehensive, especially if the probable cause may not be identified. When conducting the root cause analysis, all aspects must be considered, and a thoroughly objective approach must be taken to evaluate the cause of the deviation. Identifying the probable cause is critical. It determines the type and degree of CAPAs implemented, which affects the likelihood of the deviation recurring.
- **Determine if this is a recurrence of a previous deviation or a first-time incident.** Is there a pattern of microbial contamination? Were previous CAPAs effective?
- **Determine the potential impact.** What impact might this incident have on other batches manufactured in the facility? This impact assessment may include pharmaceutical ingredients, manufacturing equipment, or other processing lines in the same facility, other sites within the company that are manufacturing the same or related products, and distributed product.

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- **Determine the level of risk.** What risk might be posed to a person taking the product? What effect could there be on the stability, safety, and efficacy of the product over the shelf life of the product?
- **Identify CAPAs required and develop a plan for implementation.** Create a list of all the possible solutions that would prevent the deviation from recurring and select the most effective changes for implementation.
- **Clearly identify the deviation.** State what the deviation is and how it exceeded the limits. Be as specific, simple, and concise as possible.
- **Decide what, if any, manufacturing should cease during the investigation.** Depending on the nature of the deviation (e.g., high microbial counts, damage to facility, microorganisms identified), the team should consider recommending that manufacturing in the affected area should cease until remediation (e.g., extra cleaning and sanitization, more environmental sampling) has been completed and before the investigation is closed. All actions taken to demonstrate containment of the deviation should be documented.
- **Maintain clear communications.** Interview personnel involved and revise the investigation as needed. Provide routine updates to management and ensure that members of the team are well informed and kept up to date on the progress of the investigation.
- **Inspect the area where the deviation occurred.** Locate the deviation on a plan of the facility and look at surrounding areas that might contribute to the microbial contamination. Consider personnel and material flows, and nonroutine events, as well as temperature, humidity, and space pressurization excursions.
- **Look for trends.** Review prior related investigations to assess. Is there an adverse trend—same location, common personnel involvement, recurring microorganisms, root cause? If an adverse trend is identified, act promptly to remediate the problem.
- **Evaluate data.** A thorough review of production or batch records should be conducted to determine if falsification or incorrect recording of data from either intentional or unintentional human error during the time of recording results occurred ***(35)***.

While SOPs are crucial to effectively manage an investigation, other tools can provide valuable assistance in identifying possible causes of the deviation, such as brainstorming, fishbone diagram, FMEA/HAACP, cause-and-effect analysis, and the 5-why technique.

## 5.1 Root Cause Analysis

When writing a manufacturing investigation, use of root cause analysis tools are recommended ***(36)***. One common approach is the use of a fishbone diagram (**Figure 5.1-1**); it can be used to ensure that all inputs that may have contributed to the deviation are adequately considered and assessed as a potential root cause.

Thinking through the entire scope of factors that may have influenced the deviation is very important, as is recording each potential root cause identified in the investigation, even it is determined not likely to have been a contributing factor. For example, if pharmaceutical ingredients and packaging components were assessed and were determined to not have caused the deviation, the investigation should reflect that information. This demonstrates that the investigator performed due diligence in examining the potential impact of these materials as a source of contamination and subsequently ruled them out as a potential root cause.

The categories of contributing factors to be considered in a root cause analysis may vary based on the process involved. They can also be modified depending on the situation, for example, the dosage form, facility design, or nature of the deviation itself, location within the facility, sample type, classified area level, or involvement of personnel. The investigating team is responsible for ensuring that all contributing factors have been assessed so that no potential root cause of the deviation is overlooked.

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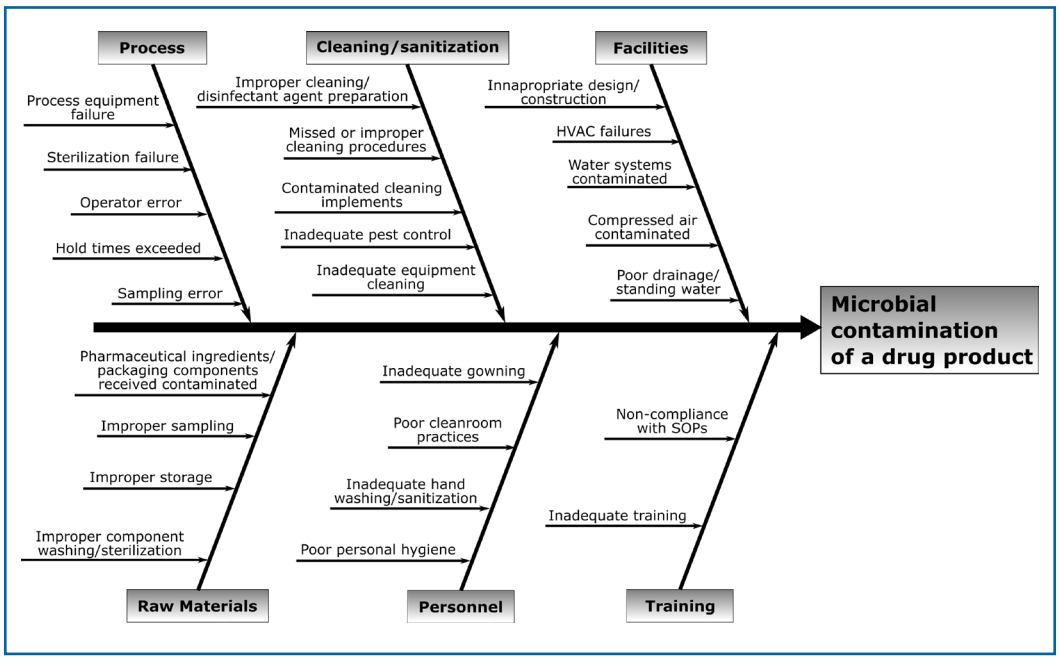


**Figure 5.1-1** Fishbone Diagram Examining Microbial Contamination of a Drug Product

In addition to the information the fishbone diagram provides, **Tables 5.1-1, 5.1-2,** and **5.1-3** present further questions which can be used as applicable to the product type manufactured to aid in the investigation.

**Table 5.1-1** Factors to Consider in Root Cause Analysis

| Area of Consideration | Questions for Consideration |
|---|---|
| Raw Materials | Have all relevant certificates of analysis been reviewed, with special emphasis on microbial testing? |
| | Have there been any recent notifications from the vendor regarding changes to their raw materials or processes? |
| | Have any of the vendor's suppliers or subsuppliers changed? Has your company been notified appropriately? |
| | If your firm comprises multiple sites that share vendors, have other sites had any complaints or issues? |
| | Did the last audit of the supplier/vendor have no microbial quality issues? |
| Personnel and Training | Have the personnel responsible for manufacturing the product been trained appropriately? |
| | Was the relevant staff (e.g., cleaning staff, engineers, maintenance workers) involved in the manufacturing process around the time of the event interviewed? Were their responses documented? |
| | Have personnel recently transferred to a new function? Did an error occur due to complexity of the function? |
| | Are all personnel trained on proper aseptic techniques? |
| | Are all personnel trained on proper cleanroom behaviors? |
| | Are personnel following procedures for operating in a classified area? |
| | Were any personnel observed to be ill or infected at the time of sampling? |
| | Was the occupancy count or time spent in the room exceeded prior to or during sampling? |
| | Were the personnel involved different from the personnel who routinely perform a particular function? |

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(Table 5.1-1 Continued)

| Area of Consideration | Questions for Consideration |
|---|---|
| Personnel and Training | How long have the personnel been performing this function? Are new employees involved? If so, are they all experienced? |
| | Have recently transferred or temporary operators been in the manufacturing area? |
| | Are the critical tasks performed infrequently? When was the last time the personnel involved performed the tasks? |
| | Are contractors (e.g., an outside cleaning or maintenance service) employed? |
| | How are contractors trained and qualified? |
| | Is operator performance evaluated on an ongoing basis (i.e., performance monitoring)? |
| | Is a particular operator or group of operators associated with a disproportionate number of deviations? |
| | Did any events occur around the time of the deviation that could have impacted operator performance (e.g., a distraction or interruption of activities, including upcoming holiday breaks; union activity; personal conflict; reorganization)? |
| | Gowning |
| | Is the personal protective equipment (PPE) appropriate for the manufacturing area? |
| | Are accessories like makeup and jewelry removed prior to entering classified areas? |
| | Is the gowning procedure clear and easy to follow? |
| | Were PPE appropriately stored? |
| | Did the operators don their gowning materials correctly? |
| | Are soiled or damaged gowns replaced routinely? |
| | Is the PPE properly stocked and organized in a way that permits ease of cleaning while not obstructing airflow? |
| | Are the operators' gloves sanitized periodically? |
| | If the area in question has video recording of personnel activities, were any deviations noted? |
| | Were personnel monitoring results acceptable for each operator? |
| Facility | Were any changes made in the area during the period in question? |
| | Are the walls, floors, and ceilings made of smooth and nonporous materials? |
| | Are all the surfaces visibly clean, free of dust, and not deteriorating? |
| | Are all windows flush mounted and properly sealed? |
| | Are all electrical outlets well covered to enable cleaning? |
| | Is there freestanding water or excess water in the manufacturing area? |
| | Are all floor drains (where applicable) clean and free of residual water and sanitized periodically? |
| | Were the temperature, differential pressure, and humidity requirements met? |
| | Were parts and equipment inspected for signs of wear and tear? |
| | Has landscaping recently been done outside the entrances to the building, particularly the application of mulch? |
| | Was there ongoing construction in the facility? Was the construction properly isolated? |
| | Are the procedures for extraneous (i.e., contractor's) equipment sanitization sufficiently robust? |
| | Air Supply |
| | Was the HVAC system in proper working order? |
| | Are the HEPA filters within their intended re-certification period? |
| | Are the HEPA filters properly installed and clean? |
| | Was any process equipment or materials obstructing proper air flow? |
| | Are the space pressurization requirements maintained? |


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(Table 5.1-1 Continued)

| Area of Consideration | Questions for Consideration |
|---|---|
| Facility | Material and Personnel Flow |
| | Is there adequate separation of "clean" and "dirty" areas? How are transitional spaces constructed and managed? |
| | How is personnel and equipment access controlled (e.g., any evidence of sharing of access badges through security checks)? |
| | Are gowning and degowning areas separate or combined? Is this appropriate for the processes being performed? |
| | Is the flow of personnel and equipment unidirectional? |
| | How are materials being sanitized when passed from one area to another (e.g., spray mist, wipe-down, immersion, autoclave)? |
| | Is the material transfer room well organized for proper flow of items? |
| | Was the material transfer room stocked with cleaning supplies? |
| | Where needed, was a transfer cart available and free of rust, dirt, etc.? |
| | When appropriate, were the materials double- or triple-bagged? |
| | Is there sufficient space in the material transfer room for personnel and for the necessary cleaning activities to take place? |
| | Where is the equipment assembled? |
| | Is the area of assembly properly controlled for this process? |
| | How are waste materials handled? Are they segregated from equipment, packaging components, and product ingredients? |
| | Environment |
| | ***Environmental Monitoring Program*** [This section applies to utilities, such as pharmaceutical water systems, clean steam, and compressed gasses, as well as the facility and process equipment.] |
| | Is the EM program appropriate for the facility and operations performed? Is the program appropriately qualified? |
| | Is the EM program risk-based? When was it last assessed? |
| | Was EM performed correctly? |
| | Are the samplers appropriately trained? Do they have a basic understanding of microbiology? |
| | ***Trending*** [This section also applies to water systems, environmental and personnel monitoring, in-process testing, and finished product testing.] |
| | Have you seen an increase in overall counts of specific genus, species, or organism types, especially a resistant type (e.g., *Bacillus*)? |
| | What isolates were recovered during sanitization validation? |
| | Was the organism identified to the correct level, i.e., characterized, speciated, or strain-typed? By which method? Is the method appropriate for the deviation? |
| | What is the probable source of the microorganism identified? Is it of human, environmental, water, or other origin? Is it part of the baseline microflora? If not, how did it enter the facility and/or product? |
| | Are your visitor/contractor training and gowning procedures sufficiently robust? |
| Cleaning and Sanitization | Was the area appropriately cleaned and the cleaning documented on and around the date in question? |
| | Was the decontamination cycle of the manufacturing isolator system completed without error? |
| | Was the cleaning/sanitizing done on a periodic basis and as defined by SOP? |
| | Were the correct (validated) cleaning/sanitizing agents used and in the correct concentration(s)? |
| | Were the correct cleaning/sanitizing tools used for the process? |
| | Was the cleaning/sanitizing equipment intact and appropriate for use? |


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(Table 5.1-1 Continued)

| Area of Consideration | Questions for Consideration |
|---|---|
| Cleaning and Sanitization | Were the operators trained on how to prepare and apply the cleaning/sanitizing agent? |
| | Were the operators gowned appropriately during the cleaning/sanitizing process? |
| | Are there known difficult-to-clean surfaces in the area(s) associated with the deviation? |
| | Have there been any recent changes to the rooms and/or to the process that would render a surface difficult to clean (e.g., physical changes, new product that leaves a residue)? |
| | Sanitization and Disinfection Efficacy |
| | Was the manufacturing facility cleaned properly and with the approved agents? |
| | Was the appropriate contact time maintained for disinfectants? |
| | Were the cleaning and disinfectant agents used within expiry? Have expiry dates been properly established for opened or site-prepared agents? |
| | Are disinfectant agents sterilized prior to use in ISO 5 and ISO 7 areas? |
| | Are the cleaning procedures and disinfectant agents validated? Have disinfectant efficacy studies been reviewed? |
| | Are these validations repeated periodically? Are newly isolated microorganisms/surfaces ever added? |
| | Can the microorganisms implicated in the deviation be killed by the validated disinfectant and sporicidal agents? |
| | If pre-wetted wipes are used for disinfecting, are the packages resealable? Do they reseal well? Have they been validated? Has an appropriate post-opening expiry date been applied (especially for IPA, which is a volatile)? |
| | Are nonfixed components and equipment routinely moved for appropriate cleaning? |
| Manufacturing Processing | Process Equipment and Instruments |
| | Has a thorough review of the following been performed? |
| | Original validation/qualification (IQ/OQ/PQ) and any updates or amendments to the original documentation? |
| | Calibration and preventive maintenance records, including any repairs or adjustments made during batch manufacture? |
| | Daily use and/or cleaning logs, as applicable, including all setup procedures and activities? |
| | Procedures for sanitization of nonsterilizable materials? Thorough evaluation of large or intricate pieces of equipment? |
| | Have the sterile boundaries been evaluated and assessed for impact, including those that supply sterile gasses or can affect product integrity? |
| | Manufacturing Process and Procedures |
| | Has a thorough review of all manufacturing processes and procedures used related to the deviation been conducted? |
| | Did the review include verification that the instructions for procedures and/or equipment use are written clearly and adequately for the intended use and full documentation of the critical steps in the batch records? |
| | Have all relevant method validation and transfer documentation, including any addendums or updated studies, been reviewed? |
| | Batch Records |
| | Have all steps been completed accurately and documented appropriately? |
| | Were all pharmaceutical ingredients added in the correct amounts, in the right sequence and time? |
| | Were all unit processing steps executed correctly? |
| | Were critical process parameters, in-process test results, and second-person reviews performed per procedure? |
| | Were nonroutine events documented (e.g., references to processing problems/difficulties, equipment change-outs, loss of environmental control)? |


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**Table 5.1-2** Root Cause Evaluation and Follow-Up

| Root Cause Analysis Portion | Considerations |
|---|---|
| Documentation | Were all equipment use/cleaning/preventive maintenance/calibration logbooks and records examined to ensure that proper procedures were followed and that all these activities were performed at proper intervals? |
| Impact Assessment | Based on the microorganism implicated and its profile, what is the risk to the patient? To the quality, efficacy, and safety of the product during its shelf life? Is the sterility assurance of the product impacted? |
| | What is the route of product administration (oral, injectable, topical, implant, etc.)? Is the patient risk high, moderate, or low? |
| | Does the product have a specific target patient population (pediatric, immune-compromised, post-operative, elderly)? |
| | Personnel monitoring: When and where does this occur? |
| | Is the microorganism likely to survive and grow in the product? |
| | Have such product attributes as preservative effectiveness, water activity, pH, and nutrient sources been considered? |
| | If the product contains antibiotic substances, has the antibiotic-resistance profile of the microorganisms isolated been reviewed? |
| | Is the microorganism likely to produce a toxin? Even if the organism is found not to cause a health risk to the patient, can it cause a bad flavor, break down the viscosity, or discolor the product? |
| | Can the organism break down the active ingredient or a component of the product and potentially affect the potency or shelf life? Could there be other post-market impacts? |
| Investigation Outcomes | What is(are) the cause(s) of the deviation? What is the level of certainty as to the conclusion?<br>Root Cause: High degree of certainty?<br>Most Probable Cause: Most likely cause?<br>Possible Cause: Likely, but not definitely, proven cause?<br>Unknown cause: Inconclusive data |

**Table 5.1-3** CAPA Determination

| | |
|---|---|
| CAPA Plan | Based on the investigation findings, has a CAPA plan been outlined to eliminate the root cause and prevent any likelihood of its reoccurrence? |
| | When determining CAPAs, have the solutions been evaluated and a consensus reached with the process owners to ensure buy-in? Are the due dates for CAPAs reasonable with as short as practically possible timelines, unless adequate justification is provided for long-term actions, e.g., capital expenditure and facility modification? Have responsible owners been assigned for each action and completion date? |
| | Have all commitments been tracked, and actions taken confirmed to ensure CAPA completion, preferably in an electronic investigation database? Have reasons for delayed implementation of a CAPA been documented? Have justification and authorization been documented for implementation extensions? |
| | How is proficiency verified by laboratory management observation and personnel monitoring? |
| | Are the alert levels and action limits in place appropriate for the area classification and/or the manufacturing process being performed? |
| | Has the effectiveness of the completed CAPAs been checked to ensure that the actions taken were successful and the deviation has not recurred? |

## 5.2 Phase II: Specific Points to Consider

### 5.2.1 Sterility-Positive Investigations

Should the laboratory confirm that a positive sterility result is valid, and no fault was attributed to the laboratory and actual testing of samples submitted for analysis, the lot of product or material should be considered nonsterile, and a comprehensive investigation into the source of contamination is required.

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Commonly, this is a collaborative effort between the QCU, laboratory, and production along with any additional SMEs relevant to the manufacturing facilities and ancillary equipment. In order to guide this step of the investigation, both scientific and regulatory considerations are provided to incorporate into your investigative strategy.

All sterility test isolates should be identified to the species level, and the data should be reviewed for any potential linkages to organisms isolated from the production environments, personnel who participate in the manufacturing process directly or indirectly by handling of the sampling containers throughout the process, and any in-process bioburden testing. If the same species of microorganism was recovered from EM or supporting activities during batch manufacturing or testing, these colonies should be identified to the strain-type level. Using the same microbial identification platform for both the sterility test isolate(s) and any isolates recovered from the production areas is advisable to eliminate potential differences in software databases if using two different microbial identification platforms. Useful regulatory guidance for conducting a sterility test failure investigation for an aseptically filled product are the 2004 FDA guidance on sterile drug products and some insightful chapters in the PDA book *Microbial Risk and Investigations* ***(3, 41, 42)***. Experience has shown that the investigational approach for aseptically filled sterile products will be significantly more complex than investigating a sterility test failure associated with terminal modalities. The following table provides a partial list of suggested questions to consider when evaluating sterility assurance data for a suspect lot of product under review during a Phase II manufacturing investigation.

**Table 5.2.1-1** Questions for Investigating Manufacturing/Process Sources Associated with a Confirmed Sterility Batch Failure

| Area of Consideration | Questions for Consideration |
|---|---|
| Sterilization of Components | Are the biological Indicators used to monitor equipment sterilization partially open and improperly stored in the refrigerator, allowing ingress of moisture to reduce spore resistance? |
| | Is equipment not held under sterile conditions after terminal sterilization? |
| | Has a new piece of equipment been added to the production train? |
| | Has there been a change or increase in the equipment pre-sterilization bioburden? |
| In-Process Bioburden Testing of Water and Product | Has there been a change in trend results for in-process control bioburden testing for product? |
| | Has there been a change to bioburden counts or the types of microorganisms recovered for water-for-injection? |
| | Is (are) the microorganism(s) recovered from the product sterility test capable of growing on the media with the incubation conditions used for production EM and personnel monitoring? (e.g., *Propionibacterium sp.*) |
| | Is the on-line sampling portal capable of being sterilized before taking a sample? |
| Batch Monitoring Data from Setup through Post-Fill | Have there been any changes in the viable EM data in lots prior to or during the suspect lot failure? |
| | Has there been any significant increase of nonviable particulate counts in the critical areas within or adjacent to the filling line? |
| | Did any critical locations (for surface or air) within the ISO 5 (Grade A) area exceed the action limit? If so, was an investigation opened to evaluate the source of contamination? |
| Personnel Interventions and Glove Monitoring in Critical Zone | Was there a new or unusual intervention for the suspect lot that has not been evaluated during the aseptic process simulation study? |
| | Do the process simulation studies include the aseptic manner for how to properly obtain in-line process samples? |
| | Is personnel or room EM actually being performed? [data integrity concern] |
| Manufacturing Batch Records | Were there any past deviations, problems, or changes to earlier batches that may have passed the sterility test that could provide an indication of the origin of the current lot sterility test failure? |
| | Are the batch records or any associated recording logs incomplete or altered that may indicate possible data integrity problems that may cover the true cause of the operation/sterility failure? |

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(Table 5.2.1-1 Continued)

| Area of Consideration | Questions for Consideration |
|---|---|
| Manufacturing Batch Records | Were there any issues with sterilization of equipment (tanks) or change parts that occurred for this batch, possibly indicating a failure in sterility assurance? |
| | Were validated holding times (e.g., mix to sterilize time) met for products that promote microbial growth? |
| | Were all other critical microbial control parameters (e.g., filter integrity test) met? |
| | For terminal sterilization processes, were all critical parameters met for the sterilization cycle? |
| | Was the calibration of sterilizer-critical instruments (e.g., temperature probes, pressure transducers, etc.) confirmed to be within limits? |
| | Were all terminal sterilization processes in a currently qualified state? |
| | Was the container closure integrity of the sterility test positive units confirmed? |
| | Were heat/sterilization process resistant bacterial spores present in the sterility test positive? If so, what the resistance value of the the contaminant compared to the biological indicator to assess the likelihood of contaminant survival after sterilization? |
| Historical Manufacturing Review | Has the firm had sterility test failures in prior product lots? What were the common factors between these product lot failures (e.g., same equipment, same personnel, same room)? |
| | Is the EM sample site map selecting the highest risk locations within the production/filling suite? |
| | Has the firm averaged EM data from critical rooms? Has the facility looked at the individual EM data before averaging? |
| Change Management | Was there any usual maintenance or equipment repairs made to the manufacturing train for the suspect lot? |
| | Do the records for HEPA-filter monitoring or repair (e.g., caulking small leaks) indicate a potential source of microbial ingress to critical workspaces? |
| | Did any storms, floods, water main leaks, or plumbing problems occur during the filling time of the suspect lot that may have been carried to the manufacturing suite? |
| Aseptic Process Simulation | Do the actual manufacturing operational practices match what is conducted during the aseptic process simulation? |
| | Was the most recent aseptic process simulation carried out within the required timeframe (i.e., 6 months)? |

Key areas for investigations related to personnel are gowning practices, operator behavior, and aseptic techniques. Personnel conducting manual aseptic operations in open ISO 5/Grade A environments (e.g. BSCs, LFHs, etc.) are monitored prior to each exit from the Grade A area via touch plates/finger dab sampling, sleeve sampling and other qualified gown locations using TSA contact and standard plates (note: sleeves and gloves should be immediately replaced with a new sterile set after each EM sampling occurs prior to re-entry back into the Grade A environment). Cleanroom operators must be trained to understand how microbial contamination can be caused by poor personal hygiene, inadequate gowning practices, or inappropriate cleanroom behaviors. Operators should be limited to ingress of hands and forearms only into the ISO 5/Grade A environments.

### 5.2.2 Determining Sources of Bacterial Endotoxin

The source of bacterial endotoxins within the sterile product manufacturing train may be transient, related to pharmaceutical ingredients, processing equipment, cleaning deficiencies, packaging components, or the pharmaceutical water system.

Areas to be considered during the manufacturing investigation in response to endotoxin contamination are outlined in **Table 5.2.2-1**.


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**Table 5.2.2-1** Endotoxin-Related Investigation Considerations

| Focus Area | Questions for Consideration |
|---|---|
| Water System and Facility | Is the water system operating in a state of control? Have there been any endotoxin excursions with the water system? Have Gram-negative bacteria been recovered from the water system? |
| | Were there any conditions within the manufacturing facilities that might favor proliferation of Gram-negative bacteria (e.g., standing water, improperly stored process equipment, pumps, hosing, or a spray bottle of water used for equipment lubrication)? |
| | Have the compressed air or gas lines been checked for the presence of water? |
| Raw Materials and Components | Were pharmaceutical excipients and drug substances properly screened for endotoxin content? Historical evidence has shown that raw materials, especially those from natural sources (i.e., animal- or plant-derived materials), may be a potential source of endotoxin. |
| | Have changes to starting materials or manufacturing processes been reported by the ingredient vendor that could impact endotoxin content of pharmaceutical ingredients? |
| | Were all product contact materials, including processing equipment and primary packaging, properly cleaned, depyrogenated, and stored to ensure they were free of detectable endotoxins? |
| Operations | Were laboratory test samples taken, transported, and stored correctly and following approved methods? |
| | Have any process deviations occurred or manufacturing changes been implemented that could have contributed to the deviation result (e.g., rinsing aseptic equipment with purified water instead of WFI, water system filters that have not been replaced on schedule)? |
| | Was the proper operation verified for any depyrogenation ovens or tunnels or endotoxin-removing columns or filtration equipment used during product handling? |
| | Were any contract manufacturers contacted regarding any changes to their processes that contact your firm's product material? |

### 5.2.3 Pharmaceutical Ingredient-Related Investigations

Pharmaceutical ingredients contaminated by Gram-negative bacteria may result in endotoxin contamination. Mold contamination could potentially lead to mycotoxin contamination. GMP requirements, including those identified in 21 CFR 210, 21 CFR 211, and ICH Quality Guideline Q7, include specific definitions as well as requirements for procedures for sampling and testing. Additionally, the regulations recommend the need for risk-based approaches to pharmaceutical ingredient procurement, testing, and inclusion in drug products ***(34, 43, 44)***.

Representative samples for microbiological testing of a pharmaceutical ingredient are taken from individual drums of the incoming shipment but the ingredient lots may not be well mixed. Creating composite samples from the top, middle, and bottom of individual drums, or taking samples from each individual drum may mask pockets of microbial contamination. Unless the warehouse sampling occurs in a sampling booth, under controlled conditions, using trained personnel with the appropriate PPE and sample containers, there is a potential for sample contamination.

With respect to the manufacturing investigation, a pharmaceutical company will probably not be able to gain full access to details of the ingredient supplier's manufacturing process or visit the manufacturing site to conduct an investigation, unless it is a requirement in the quality agreement. Many pharmaceutical excipients are viewed as commodities that are supplied to other industries without GMP compliance. Many excipient suppliers set their specifications based solely on the relevant USP monograph requirements, which may not include microbial requirements. In addition, excipient suppliers are largely unresponsive to the GMP requirement to exclude objectionable microorganisms from nonsterile drug products ***(24, 45, 46)***. **Table 5.2.3-1** provides a list of additional questions to consider during a manufacturing investigation.

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**Table 5.2.3-1** Additional considerations for raw material investigations

| Raw Material Component | Considerations |
|---|---|
| Format/Storage | What size is the bulk container? Is it sampled top, middle, and bottom? |
| | Is it a synthetic or mineral-, plant-, or animal-derived ingredient? |
| | Is it provided for sterile or nonsterile drug product manufacture? |
| | Is the raw material presented as a powder or liquid? |
| | Does it contain a preservative? If so, are preservative effectiveness tests conducted or is an assay performed? |
| | Was the raw material properly released from quarantine? |
| | Were the temperature and humidity conditions maintained during shipping and storage? |
| Vendor/Source | Have there been any recent changes in suppliers or manufacturer? |
| | Was the vendor qualification program followed? Are the audits current and passing? |
| | What is the source/origin? Plant, animal, or biological materials (chemically undefined) are often at a higher risk for contamination. |
| | What is the history of the material? Are the results trended? Has a drift from the mean been detected? |
| | Have supplier and manufacturer risk assessments been completed based on the intended use? |
| | Is a skip lot program in place? If so, should it be suspended? |
| | Has the vendor's certificate of analysis tests been verified on an annual basis? |
| | Are the manufacturer's specifications in compliance with compendial monographs? |
| Sampling | Was the sampling plan followed upon receipt? |
| | Were samples composited? Composited samples can potentially dilute the final results. |
| | Is there a pharmacopeial monograph? |
| | Is the sampling hood or room classified? Or controlled but not classified? |
| | Were the proper sterile sampling devices and containers used? |
| | For sterile processes:<br>– How are samples obtained?<br>– Are there side samples, or is the material sampled aseptically through a septum? |
| | Were the established test frequencies/test-on-expiry maintained? |
| | Are there reserve samples of the material or unused stock materials available for microbiological testing? |
| | Are samples representative of the container/lot or could there be potential pockets of contamination in the bulk material or hot spots? |

As the deviation investigation progresses, answering these or similar questions can be useful in finding the root cause. These questions can also be used in a product lifecycle approach for new raw materials, or to build a proactive continuous improvement tool to manage contamination risk.

### 5.2.4 Evaluation of Confirmed Mycoplasma Contamination

Investigation should be conducted in a timely manner as mycoplasma contamination can interrupt production and prove costly to the firm.

Common sources of mycoplasma contamination are master and secondary cell lines, raw materials, and growth medium that has not been rendered sterile using a 0.1 µm filtration step. Attention should be given to investigating these materials. The authentication, acquisition, banking, passage, and cultivation history of the master cell bank and working cell lines should be reviewed, with particular emphasis on certificates of analysis, certificates of origin, and the reported mycoplasma test results. Testing additional samples from the working cell bank may be advisable. Human contact or contaminated cell culture media and reagents may contaminate cell banks, and a new working cell bank would need to be established in order to continue production.


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Cell culture media is usually made mycoplasma-free by heat treatment, irradiation, antibiotic use, or 0.1 µm retention filtration. That these procedures are fully validated and documented should be verified. The media, equipment sterilization, and handling of inoculum production must be as well controlled as large-scale cell culture. The manufacturing batch records should be reviewed thoroughly, especially noting manufacturing deviations and staffing changes. If available, retained production materials could be tested using a rapid method to trace the potential source of the mycoplasma contamination ***(29)***.

When the mycoplasma are of human origin (*Mycoplasma orale, M. fermentans,* and *M. hominis)*, human contamination may likely play a role in the failure. Additional consideration should be given to species of mycoplasma that are of animal or nonanimal origin (*Acholeplasma laidlawii, M. hyorhinis, M. synoviae,* and *Spiroplasma*) as they may be indicative of contamination of specific growth media or additive growth components ***(46)***.

### 5.2.5 Component (Container Closure) Investigations

Some of the major causal factors of microbial contamination in primary packaging components that threaten the sterility assurance of sterile drug products include the container (e.g., vial, cartridge, flexible container, or syringe) and closure (e.g., syringe cap and plunger, stopper, seal). Specific microbial tests have the potential to fail as a result of a defective container or closure system, such as the bioburden test, sterility test, process simulation media fill (turbid vial), container closure/seal integrity test, and BET. Test failures can occur during initial component validation or revalidation testing, component release testing, in-process monitoring, finished-product release testing, or stability testing. The severity of the microbial test failure may depend on where the container–closure failure occurred. In every case, a thorough review of the components, filling, and closure operations should be performed. Involving the supplier early in the investigation is the best plan, as they can help provide data from the manufacturer. A complete history will be needed that covers manufacture supply, storage, preparation, sterilization, use, and final finished presentation in order to successfully complete the investigation.

#### 5.2.5.1 Pharmaceutical Containers

Vials, cartridges, and syringes used for sterile products are regarded as primary containers. Each presentation has unique characteristics that must be considered during a microbial failure investigation. Upon container receipt, a 100% inspection is not possible, both because of the destructive nature of functional testing, the size of the incoming shipments, and the general reluctance to return quality samples to inventory. Thus, acceptable quality-level attribute inspection plans are widely used for packaging components. After the initial testing of packaging components for bioburden, endotoxin, and physical characteristics, a defect inspection and certificate of analysis review is performed. Dimensional tests are important because the necks of containers and closures form the barrier or seal that prevents microbiological and other contamination ingress. Containers that are out of round, have seam or form defects, or have cracks can present paths of ingress. Molded glass can be traced to a specific mold at the factory to identify recurring defects. The initial concern is the physical state of the shipping containers, so an inspection for gross damage should be performed. Shrink wrapping trays of containers will stabilize the shipment, prevent damage, and reduce particulate contamination. Shipping cartons that are received infested, wet, crushed, or torn should be rejected. Containers used during production of a product with a failed microbial test should be reviewed.

#### 5.2.5.2 Pharmaceutical Closures

The most common closure for a sterile injectable product filled in a glass vial is an elastomeric stopper and crimped aluminum seal. Syringes most commonly include a shielded needle and are sealed using the rubber septum in the syringe plunger without a crimp seal. For cartridges, a combination crimp seal is generally employed. Each presentation has unique challenges but, generally, the investigation should follow the same path as that for containers. Special consideration should be given to the semipermeable nature of elastomers. A review of the sterilization process validation also may be useful.

Containers and closures can be purchased raw and unprocessed, ready-to-sterilize, and ready-to-use. These different preparations should be considered during an investigation.

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#### 5.2.5.3 Component Contamination

Post-manufacture microbiological and environmental contamination can include nonviable particles, microorganisms, and bacterial endotoxins. Viruses are generally not a concern with nonanimal raw material used to manufacture packaging components. In most cases, the high-heat extrusion process associated with glass and elastomer manufacture is microbiologically unfavorable; however, environmental controls in the manufacturing plant are often inadequate to maintain these conditions during sorting and packaging for shipment. Downstream processing is required and must include rigorous processing controls.

#### 5.2.5.4 Container–Closure Integrity

Physical deformation, damage, and out-of-tolerance dimensional specifications of both containers and closures can result in sterile products becoming nonsterile through microbial ingress. Emphasis should be given to manufacturing set-up, in-process controls, and container–closure testing at release and in stability programs. Failure of the container–closure system and the inability to maintain microbial integrity should be fully investigated.

### 5.2.6 Environmental Monitoring

The pharmaceutical industry has many controlled and/or classified environments in which products are manufactured, and the key to the success of an EM investigation is understanding these requirements and the possible sources of contamination. Out-of-level monitoring results can come from the following types of environments:

- Classified cleanrooms for sterile product manufacturing
- Controlled or classified rooms for nonsterile product manufacturing
- Controlled or classified rooms for low bioburden product manufacturing (e.g., biologics)
- Laminar flow hoods, restrictive access barrier systems (RABS), and isolator systems used for product manufacturing or finished-product release testing

This section will cover the potential sources of contamination, and points to consider when investigating out-of-level EM results. Eliminating the potential sources of contamination of the manufacturing environment must be established by risk assessment during facility and program development. Measures for control, however, can be implemented based on the following key concepts:

- Minimize the entry of contaminants by incorporating good facility design, proper layout, efficient material and personnel flow, and sound supplier controls
- Reduce contamination during the manufacturing process using well-developed process and equipment controls
- Establish and maintain proper hygiene and PPE practices
- Institute appropriate aseptic techniques and cleanroom behaviors
- Assure a state of control through routine EM and analysis

Possible sources of contamination that may enter the manufacturing or testing environment are summarized in **Table 5.2.6-1** ***(48, 49)***.

**Table 5.2.6-1** Potential Sources of Contamination

| Source | Example |
|---|---|
| Pharmaceutical Excipients | Corn starch, dicalcium biphosphate, purified water |
| Air Supply | Unfiltered air, turbulent air, mixed air |
| Building Materials | Plastic, sheetrock, wood, metal, epoxy |
| Equipment | Parts washers, hoses, air samplers, pallet jacks, transfer carts |
| Disinfectants | Preparation, expiration, sterile (where appropriate) |
| Containers | Storage totes, waste receptacles |
| Packaging | Cardboard, gowning supplies, paper |
| Supplies | Mops, wipes, labels, booties |


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#### 5.2.6.1 Historical Data Evaluation

Reviewing historical data for adverse trends can be helpful to the investigation. Trending will consider sample type, organism type, test type, classified area, operational shift, or seasonal pattern. Trending can be both quantitative (i.e., count per unit surface area or air volume) or qualitative (i.e., frequency of exceeding an action limit), as illustrated by these specific examples:

- Predefined number of consecutive alerts, actions, or contamination rates exceeded
- Same microorganism or related microorganisms isolated across multiple locations, monitoring types, or rooms
- Mold or Gram-negative or Gram-positive spore-forming organisms recovered in the critical locations associated with a specific manufacturing process

#### 5.2.6.2 Manufacturing Process Evaluation

The purpose of an EM program is to determine the adequacy of the environmental controls and the aseptic technique of personnel in the manufacturing area. A well-maintained and well-operated facility requires good facility design, operation, and maintenance; functional HVAC systems; temperature and humidity control; supply of HEPA-filtered, laminar flow air; proper movement of materials, equipment, and personnel; and a sound facility cleaning and sanitization program. If these requirements are not maintained, control of the environment will be lost, and an EM excursion will occur. The microbiologist on the investigation team must be experienced in EM and understand manufacturing operations. Questions commonly asked during the manufacturing portion of an EM investigation are outlined in **Table 5.1-1.**

Loss of environmental control in the cleanroom should be considered a potential source of microbial contamination. Excursions in temperature, differential pressure gradients, humidity, nonviable particulate levels, and appropriate cleaning and disinfection activities can contribute to microbial contamination in the cleanroom environment.

Microbial contamination can enter the manufacturing facility, including cleanrooms, in a variety of ways; in fact, the most common cause of fungal spore and bacterial endospore contamination in cleanrooms are items that enter the cleanroom. Some customary items include tools, processing equipment, communication devices, boxes, carts, cleaning and sanitizing equipment, and batch records. Cardboard and other cellulose-based materials known to shed particles should never enter GMP-compliant manufacturing areas. Items entering critical aseptic processing areas should first be double-wrapped, unwrapped, and wiped down with a sporicidal agent.

The age and maintenance level of facilities can affect contamination rates in the cleanroom ***(50)***. Ceiling tiles, HEPA-filter units, cleanroom flooring, materials of construction, drainage systems, and duct work are areas of concern in aging facilities that may need to be kept on a routine schedule for repairs.

Drains in manufacturing facilities require routine disinfection with a sporicidal agent to prevent microbial proliferation.

Examples of CAPAs to remediate microbial contamination in the manufacturing facility are outlined in **Table 5.2.6.2-1**

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**Table 5.2.6.2-1** Examples of Corrective Actions

| System | Example of Corrective Action |
|---|---|
| Facility | • Review risk for product or component contact<br>• Inspect preventive maintenance records<br>• Reduce equipment or materials in the location<br>• Upgrade facility surface finishes, i.e., floors, doors, walls, and ceilings |
| HVAC System | • Increase the number of air exchanges in the area<br>• Lower temperature and humidity settings<br>• Repair or replace HEPA filters or housing grates<br>• Add low wall air returns to eliminate particles |
| Personnel | • Reduce the number of operators in confined areas<br>• Reduce the number of operator interventions<br>• Interview production staff for area improvement suggestions<br>• Retrain or requalify operators with improved techniques |

### 5.2.6.3 Resampling or Investigational Monitoring

The need for additional monitoring to determine root cause (to confirm or exclude hypotheses with regards to the root cause) should be considered early in the investigation to avoid losing time and to ensure that microbiological results will be representative of the conditions.

Based on the type of manufacturing facility, the frequency for routine environmental monitoring should be considered and a decision should be made as to whether resampling or investigational monitoring is warranted. Scenarios in which resampling may be useful include low frequency monitoring locations and locations found as important during facility walk-throughs. If the isolate identified as part of an EM excursion or routine sample is considered a unique microorganism (e.g., first time recovered in the facility), some investigational monitoring may be warranted. At the discretion of the QC microbiologist and in conjunction with approval from the QCU, additional locations may be sampled, or specialized microbiological media may be utilized. For example, recovery of mold from the wall, but not the floor, may suggest that sampling be conducted of nearby ceiling locations, adjacent walls, and additional portions of the same wall, in order to determine the extent of the fungal colonization in the room. Specific microbiological medium that may improve recovery and/or detection of the unique microorganism(s) may be used in addition to the standard monitoring medium (e.g., Sabouraud dextrose agar may be useful in response to mold recoveries). SOPs should allow discretionary sampling that may aid the investigation.

## 5.2.7 Investigation of Confirmed Contamination During A Process Simulation/Media Fill

Microorganisms isolated from turbid vials may originate from cleanroom personnel, equipment surfaces, walls and ceilings, floor drains, packaging materials, disinfectants, or materials transported into the aseptic filling area. They may correspond to microorganisms isolated during EM and product sterility testing. **Table 5.2.7-1** contains a sample of representative microorganisms that, from task force members' experience, have been found in failing media fills and their possible origins. These microorganisms were benchmarked against a recent review of cleanroom microorganisms ***(40)***.


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**Table 5.2.7-1** Possible Origin of Representative Microorganisms

| Possible Origin of Microorganisms | Bacteria | Yeast and Mold |
|---|---|---|
| **Cleanroom Personnel** | *Micrococcus luteus, M. lylae, Staphylococcus epidermidis, S. hominis, Corynebacterium tuberculostearicum, S. capitis, S. warneri, S. hemolyticus,* and *Cutibacterium acnes* | *Candidia albicans, C. parapsilosis, Rhodotorula rubra, Torulopsis spp., Trichosporon cutaneum, Microsporum gypseum, Trichophyton rubrum, Alternaria spp., Furarium spp., Paecilomyces spp., Aspergillus flavus* and *Penicillium spp.* |
| **Laminar Flow Air and Compressed Gases** | *Bacillus spp., Paenibacillus spp.* | *Alternaria spp., Aspergillus spp. (A. vesicular, A. niger, A. fumigatus), Cladosporium spp., Fusarium spp.* |
| **Filling equipment** | *Bacillus spp., Paenibacillus spp.* | *Alternaria spp., Aspergillus spp. (A. vesicular, A. niger, A. fumigatus), Cladosporium spp., Fusarium spp.* |
| **Facility Curtains, Walls, and Ceilings** | *Bacillus spp., Paenibacillus spp.* | *Alternaria spp., Aspergillus spp. (A. vesicular, A. niger, A. fumigatus), Cladosporium spp., Fusarium spp.* |
| **Packaging Material** | *Bacillus spp., Paenibacillus spp.* | *Penicillium spp., Aspergillus spp.* |
| **Improperly Made Disinfectants** | *Burkholderia cepacia, Pseudomonas putida/ fluorescens,* and *Flavimona oryzihabitans* | *Alternaria spp., Aspergillus spp., and Cladosporium spp.* |
| **Materials, Totes, and Carts Transported into the Aseptic Filling Area** | *Bacillus spp., Paenibacillus spp.* | *Alternaria spp., Aspergillus spp., Cladosporium spp., and Fusarium spp.* |

The location of the turbid vial(s)within the time sequence of the media fill may aid the investigation. Was the turbid vial from the beginning, middle, or end of the filling operation? A turbid vial from the beginning of the fill may be the result of microbial contamination during aseptic assembling of the filling pistons or needles at the filling station; from the middle of the fill, it may be associated with an operator intervention, like clearing a conveyer belt of downed vials, conducting a periodic fill-volume check, or replenishing the stopper hopper; from the end of the fill, it may be associated with poor aseptic practices or a result of operation fatigue. If filled units are numbered sequentially or time-stamped, it may be possible to review the video of the media fill and determine if an event or intervention occurred and when the individual media fill vial was contaminated. If the media fill cannot be videotaped, an observer should log all events and interventions that occurred to aid an investigation or improve cleanroom practices.

## 5.2.8 Utilities Monitoring

If the initial laboratory investigation (**Section 4.0**) does not find a conclusive source of a laboratory error that explains the deviation, deviations related to pharmaceutical water systems should be examined.

Water is used in many ways in pharmaceutical manufacturing, and the required grade of water varies depending upon how it is used. Compendia define the minimum quality attributes that each grade of water must meet and may specify its method of preparation. Pharmaceutical water systems consist of several components, including a water supply, a robust purification system that should consistently produce water of the intended quality under a range of conditions, and the storage and distribution systems that assure the water continues to meet all required quality attributes until it is delivered to the process. Together, the design, installation, and qualification of the water system, as well as the procedures for its operation, use, and maintenance, ensure the system consistently performs as intended and in a state of control.

The cross-functional team charged with investigating any deviation related to a water system should be familiar with the potential risks associated with its design, installation, operation, use, and maintenance. This team also needs to understand both the value and the limitations of data derived from system qualification and monitoring.

A monitoring plan should define the locations and frequency of sampling and the attributes for testing. The plan should encompass the entire system, from the water supply to the points of use, and relevant points in between on the generation, storage, and distribution systems. The sampling plan should reflect the results of intensive sampling during qualification, recent historical data, risk factors,

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and mitigations. Monitoring points should be carefully selected to ensure they provide the intended information regarding control of the system, including the information needed to support the initial investigation of a deviation at that point or at other points on the system.

Routine monitoring of the water generation, storage, and distribution systems verifies that they continue to operate in a state of control, while monitoring at the points-of-use evaluates the quality of water delivered to the process. Points-of-use should be regularly sampled and tested (typically off-line) for chemical attributes. These chemical measures can be a useful reference when conducting an investigation of a deviation. Measurements taken before and after unit operations, such as carbon beds, deionizers, UV lights, and reverse osmosis system, can provide data to assess whether they continue to function as expected.

Microbiological monitoring typically involves manual sample collection from a storage tank, a site on the loop, or at a point of use; the transport, storage, and analysis of the sample; and, ultimately, the recording, interpretation, and reporting of results. Each step should be carefully designed to minimize potential errors and to ensure the reported result accurately reflects the condition of the material sampled. In addition, several monitoring systems are available for at-line monitoring of microbial attributes and, as with other instruments for the sampling and measurement of critical attributes, these should be properly qualified. For additional guidance on monitoring plans, refer to PDA *Technical Report No. 13 Fundamentals of an Environmental Monitoring Program* ***(4)***.

The approach to investigating the root cause of a deviation on a pharmaceutical water system should be guided by the location and timing of the sample and by the test involved. If multiple deviations of the same type and level are observed, a cause that affects the entire system is more likely. If the deviation is not consistent with results from other contemporaneous samples, causes related to the sample point or to sampling or testing may be suspected. Each water system deviation is unique, and its investigation may consider factors related to the design, installation, operation, maintenance, change history, and use of the system. The monitoring history, including past deviation investigations, related CAPAs, and typical microbial flora of the system, will provide useful references.

In the case of a deviation for bioburden, the identity of the organism recovered is essential information. Recovery of organisms that thrive in water and form biofilms may suggest one set of causes, whereas recovery organisms that are not compatible with the water system may suggest another set of causes, possibly related to environmental contamination.

Potable water is the starting ingredient for both purified water and WFI. Standards set by regional authorities, at minimum, require potable water to be free of coliforms and to contain bioburden at a specified level (e.g., ≤ 500 CFU/mL per USP <1231>), and they may define additional requirements. Purified water must meet the compendial limits for conductivity, total organic carbon, and bioburden (≤ 100 CFU/mL). WFI must meet the same physical and chemical criteria, as well as more stringent microbiological criteria for bioburden (< 10 CFU/100 mL) and bacterial endotoxin (< 0.25 EU/mL).

The source and method of delivery of potable water can introduce significant risks to control of the water system. Potable water quality may be affected by seasonal variations due to runoff of melting snow or periods of low precipitation. Agricultural or industrial runoff may enter surface water sources or aquifers. An aging infrastructure may increase the chance of contamination entering the water supply via a damaged water main ***(50)***. Likewise, water delivery via truck introduces a variety of risks that may vary from day to day and truck to truck. The sampling and analysis plan for the potable water supply should consider the risks related to the source and method of supply and should consider, in advance, the information needed to investigate a deviation found in a sample from the potable water supply or from the system downstream.

Risks related to the design and installation of the water system should be identified and remediated prior to qualification testing. System boundaries should provide an effective barrier to microbial con-

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tamination, including microbial retentive filters on holding tanks and the use of appropriate elastomers. Dead legs, as well as poor welds and other irregular surfaces, increase the risk of biofilm development. Recent design standards specify that pipes branching off the main recirculation loop should not exceed 1.5 times the pipe diameter, and the use of valves with no dead legs is now common.

Investigation of a deviation should include a review of system operations, including control of physical parameters, and results of nonmicrobiological monitoring of the system. Operational controls for storage and distribution systems typically include parameters such as temperature, flow rate, and pressure, and can show whether the system remained flooded or flowing and at the required temperature. Additional parameters may be defined for pretreatment stages such as carbon beds, deionizers, and reverse osmosis systems. On-line, at-line, and off-line measurements of physical and chemical attributes such as hardness, conductivity, and total organic carbon, provide near continuous data on chemical attributes. They also allow prompt notification of an unexpected shift in performance, which can be helpful in investigating deviations. The rate of water usage affects the stagnant time of water in the system and may be related to the root cause.

The investigation of a deviation may include a review of routine and corrective maintenance records. Invasive maintenance adds contamination risks, so measures to mitigate these risks and return the system to service should be prepared in advance, whenever possible. Water system maintenance should ensure the system remains integral and continues to operate in a state of control. A well-designed program assures that the system boundaries continue to provide an effective barrier that prevents microbial contamination. Measures should include periodic replacement of key parts, such as elastomers and vent filters on holding tanks, as the failure of these elements can open the system to the external environment. The investigation of a deviation should also consider the potential for wear and damage to durable parts such as piping. Pipes may thin over time, and damage can occur due to unplanned events such as hydraulic shock. Water leaks should be repaired promptly to reduce the opportunity for contamination; however, the investigation of a deviation should recognize that not all leaks, a small leak of cooling water through a cracked heat exchanger tube, for example, are easily detected.

Practices for operation of the sample or use port should be considered, especially if the deviation is not consistent with other results. Flushing prior to sampling (and use) is important to physically remove the opportunistic organisms found on hoses, sampling apparatus, or the sample port. Use procedures should ensure that dead legs are not created through the use of valves between the water system and points downstream. Hoses should be removed when not in use and should be stored in a free-draining position.

#### 5.2.8.1 Compressed Gasses

Compressed gasses used during the manufacturing and testing of pharmaceutical products are air, nitrogen, oxygen, and carbon dioxide. Inappropriate design and piping for distribution of these gaseous materials can introduce unwanted microorganisms. Pharmaceutical-grade compressed air is treated to remove moisture, and chemical residue like lubricants and point-of-use sterilizing-grade filters are employed. The introduction of uncontrolled water moisture or inadequately filtered compressed gas, coupled with poorly designed pretreatment and sanitization processes of the gas delivery systems, can adversely affect the system and lead to product contamination. The types of methodologies used for testing the microbial presence within these systems are important because when finished pharmaceutical products are found adulterated, the eventual investigation by the QCU will need to find the source of the contamination or, at least, eliminate the possibility that it was from a compressed gas.

Even though these utility systems have been qualified to perform as designed, the materials used during their construction may contain unexpected component defects (i.e., poor welding or incomplete passivation), as well as presenting potential ingress of grease and oils used for equipment lubrication. In addition, the metals and plastics may deteriorate over time, causing breaches or leaks in the plumbing materials or in components connected to the pipes (e.g., gauges, shut-off valves).

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# 6.0 Conclusions

The investigation of microbial monitoring and testing deviations is a critical activity in the pharmaceutical industry that is not adequately covered by the FDA *Guidance for Industry: Investigating Out of Specification (OOS) Test Results for Pharmaceutical Production* ***(1)***. This technical report provides industry best practices related to microbial investigations to eliminate that gap in guidance.

These investigations are operationally divided into two phases—Phase I laboratory investigations and Phase II manufacturing investigations—with different teams responsible for each activity. Because of the importance of timeliness in microbial investigations, the manufacturing investigation may be initiated prior to completion of the laboratory investigation.

The objective of the laboratory investigation is to confirm that the sampling and testing was conducted correctly, and the reported result was valid. The investigational team is usually limited to the microbiologist conducting the test, the laboratory management, an SME with knowledge and experience with the test, and a representative of the QCU.

The major points to consider in a laboratory investigation are:

- Integrity of the test sample used in the microbial test
- Educational background, skill, and experience of the microbiologist conducting the test
- Testing and environmental and personnel monitoring history associated with the microbiologist
- Suitability of the equipment, supplies, media, diluents, and reagents used for the test
- Environmental controls, including microbial testing facilities, laminar flow hoods, isolators, and PPE associated with the test
- Reading of plates, scoring of microbiological broth, recording of data, accuracy of calculations, and review and reporting of the results
- Microbial identification of the isolate implicated in the deviation

Once the validity of the microbial test is confirmed, a multifunctional team is assembled to conduct the manufacturing investigation related to the failure. Representatives of the microbiology laboratory, manufacturing, engineering, maintenance, and process validation may be members of the manufacturing investigation team usually led by the QCU. To increase the chances of a successful investigation, an experienced pharmaceutical microbiologist familiar with manufacturing operations is usually added as an SME.

The major points to consider in a manufacturing investigation are:

- Unit operation and location within the facility directly related to the failure
- Possible contribution of surrounding support areas
- Possible contribution to the failure of deviations in environmental controls, i.e., space pressurization, airflow patterns, temperature, humidity, power outages, material flow, facility cleaning, and sanitization
- Possible contribution of pharmaceutical ingredients, ingredient water, in-process materials, packaging components, and utilities
- Deviation in critical manufacturing steps, as documented in the batch record
- Recent change controls
- Related in-process bioburden, environmental, and personnel monitoring deviations
- Interviews with operators, laboratory management, and production managers as to the possible cause of the failure
- Follow-up microbial monitoring and testing

The investigation reports must adequately document the team findings. If the root cause of the out-of-limit or out-of-specification microbial test result cannot be definitively established, the most probable cause or causes must be presented. Investigating out-of-trend results may help identify laboratory and/or manufacturing issue before they become out-of-specification results. To prevent the recurrence of these microbial failures, a CAPA plan must be implemented, with responsible individuals reporting their progress on the plan to the QCU and their management.

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